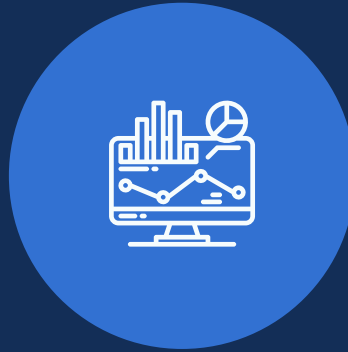




Data Visualization

Defining the Purpose



Define the purpose and audience for your visualization.



Select the appropriate visual to tell the story or communicate the message.



Focus attention on the key message using data viz best practices.



Design professional and accessible visuals and dashboards to build trust.



Create meaningful dashboards that tell stories and inspire decision making.



Identify examples of bad data visualization and dashboarding.

Purpose & Context

Purpose & Context

1



Establishing Purpose

Setting the groundwork for effective data visualization.

2



Focus Questions

What is the main question driving the visual.

3



Understanding Context

Know the importance of the data to the business, who the audience is, and the context behind the focus question.

4

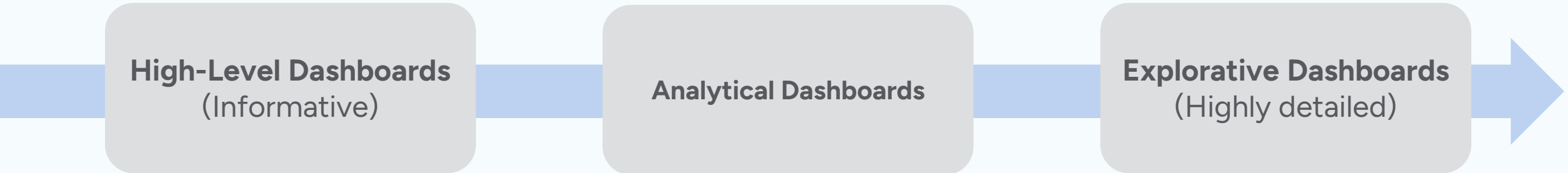


Importance of Purpose and Context

The compass guiding all decisions from visual to ensuring clarity of the outcomes in visuals and dashboards.

Defining the Purpose

A dashboard should have a focus question

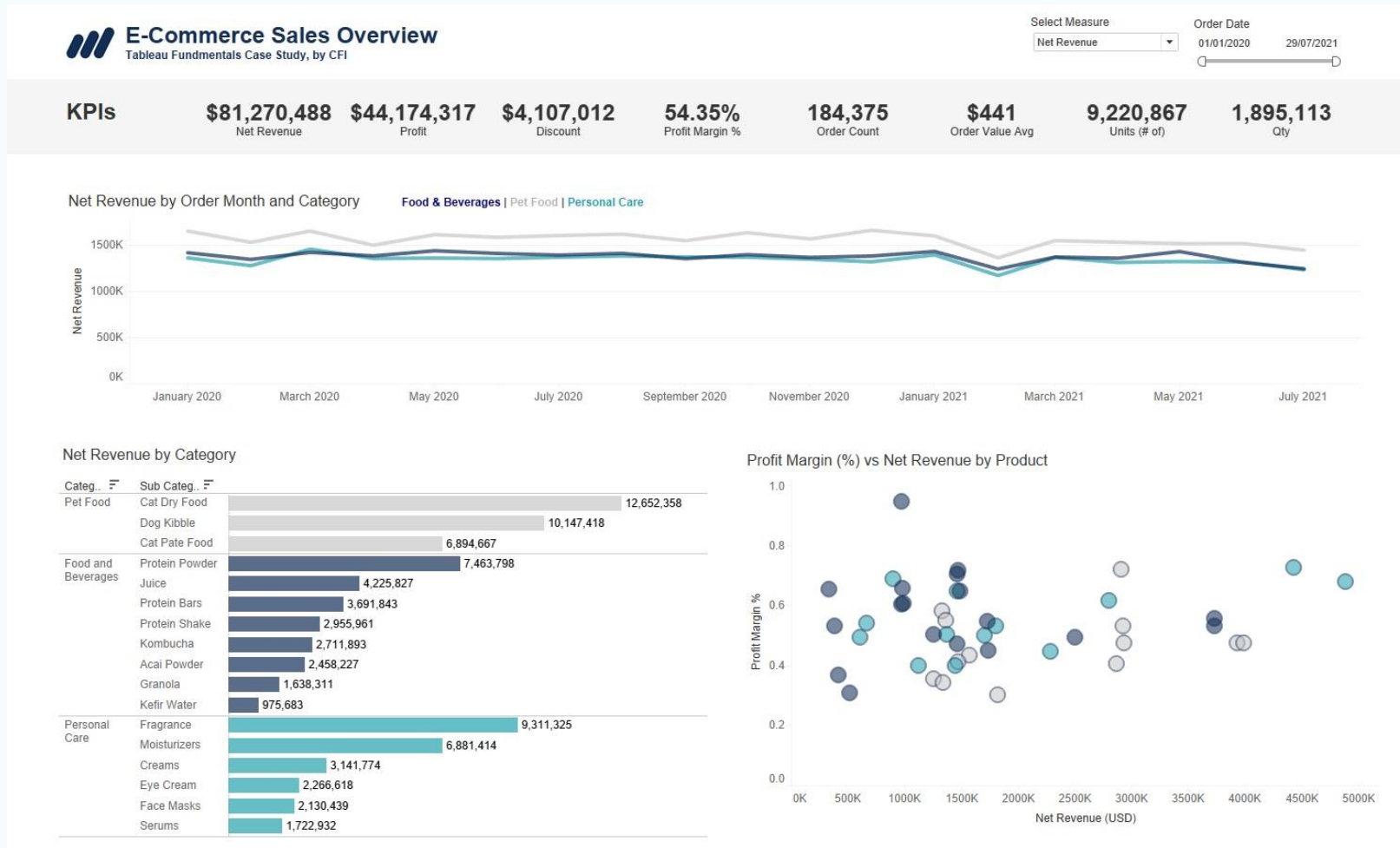


High-Level Dashboards
(Informative)

Analytical Dashboards

Explorative Dashboards
(Highly detailed)

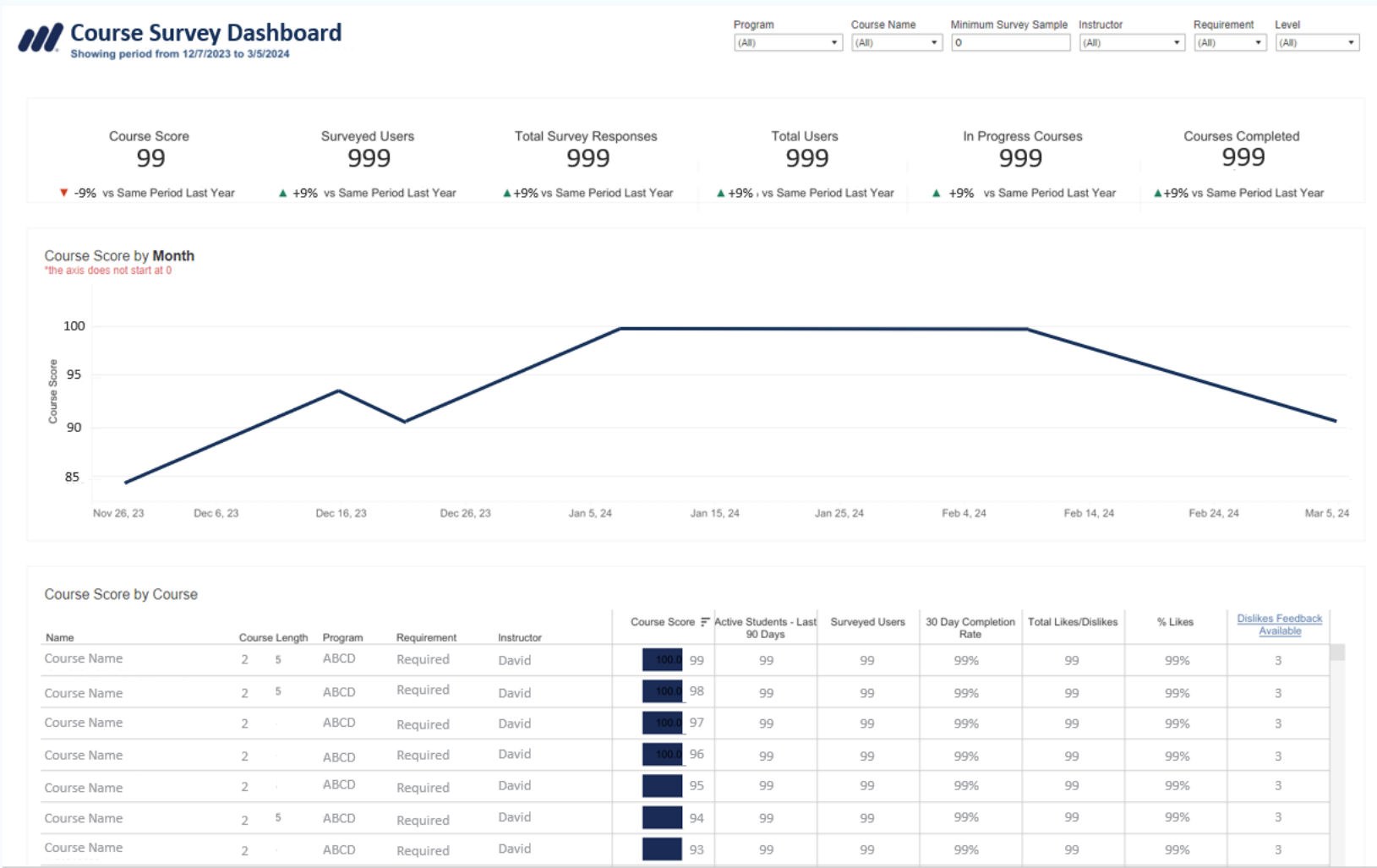
Types of Dashboards – 1. High-Level Dashboards



1. High-Level Dashboards

- Communicate key metrics or trends
- One version of the truth
- Simplicity (fewer visuals, less filtering)
- Top-level summaries are the key
- Used by senior leaders

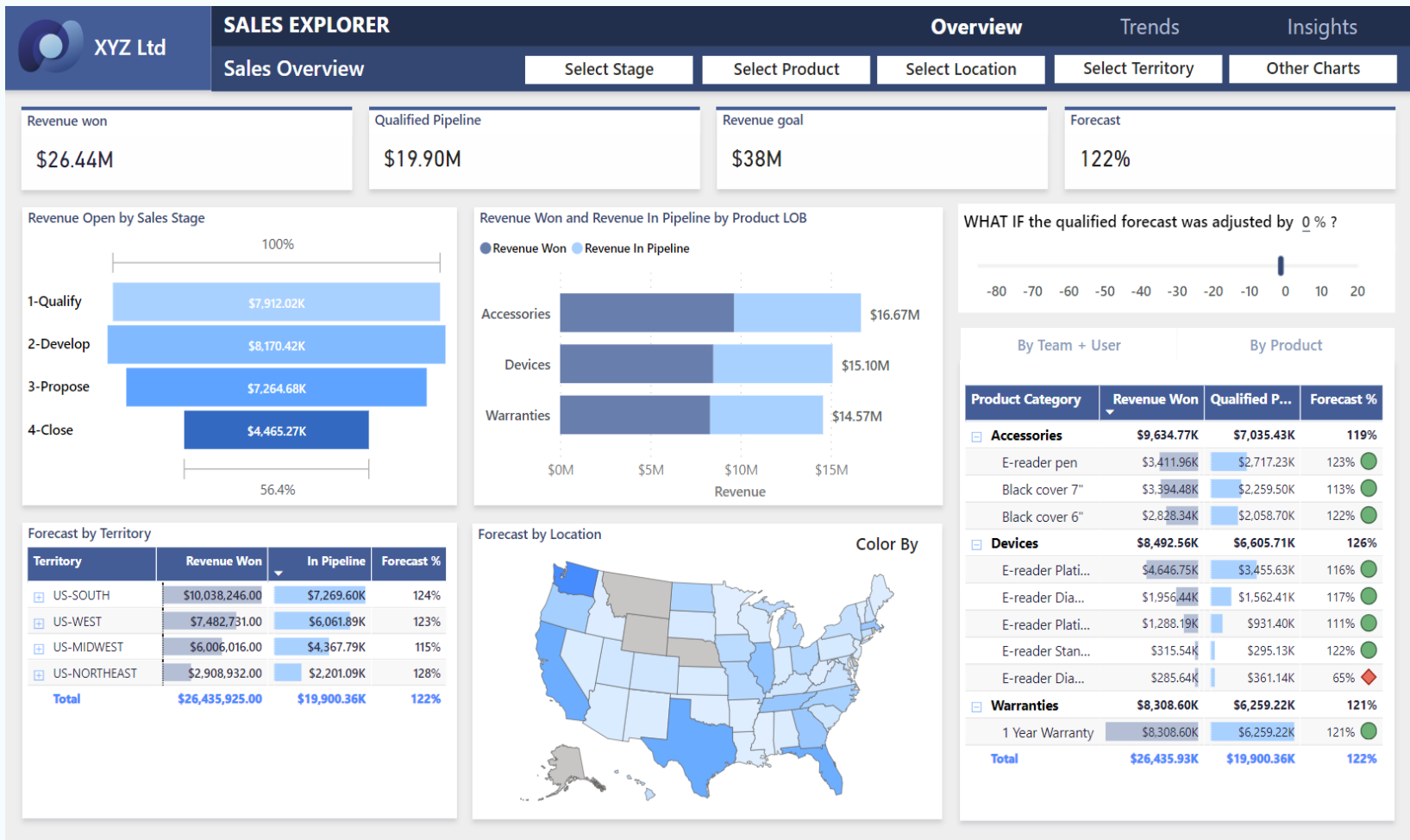
Types of Dashboards – 2. Analytical Dashboard



2. Analytical Dashboards

- Answer a specific question or problem
- Perhaps help identify a root cause
- Allow us to dig a little deeper with a few filters
- Used by managers and senior analysts

Types of Dashboards – 3. Explorative Dashboards



3. Explorative Dashboards

- More granular data for knowledgeable analysts who know the subject well.
- Many opportunities to slice and dice, to explore insights and trends and individual transactions.
- Not relevant for senior managers and execs.

Presenting to Persuade



- What is the key message?
- Are there any obvious questions raised?
- Have they been answered clearly?

Types of Dashboards – Summary

Key Learning Points

- **Purpose steers design and presentation.**
- **Simplicity is key** for informational views – everyone gets the same views.
- Analytical dashboards require **clarity** in patterns and trends, which is key for exploration.
- Exploratory dashboards require a **well-organized** layout and navigation.
- **Persuasion requires a compelling narrative** with a clear key message.

Key Questions to Ask

- Why are we creating this dashboard?
- What question are we trying to answer?
- What level of detail is appropriate?
- What level of interactivity is required?

Knowing Your Audience

Imagine you're a roller coaster designer and engineer. You've designed a new roller coaster and are sharing the design with...

An Engineer Colleague

Information Shared

- Grade of Steel
- Severity of Turns
- Ride Time
- Safety Tolerances



Your 6-Year-Old Child

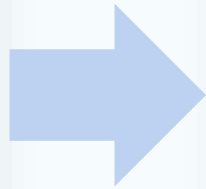
Information Shared

- Rollercoaster Color
- Number of Loops
- Volume of Screams

Knowing Your Audience

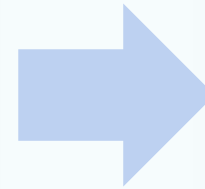
Know Your Audience

- Are they experts in the field?
- Are they beginners?



Craft Your Message

- Adjust visuals, language, and depth based on the audience



Tailor Your Presentation

- Match visuals and content to audience preferences

Setting the Context

Setting the context involves providing your audience with the necessary background information to fully understand your visuals. This may be needed in both an in-person or dashboard-only presentation.

Prior Knowledge

- Does your audience have a deep knowledge of this domain?
- Does your audience understand domain-specific terminology?

Data Quality

- Where did the data come from?
- Are there any reasons to doubt its reliability?

Date Range

- Over what period were these metrics measured?
- What periods are we comparing to?

Setting the Context

Setting the context involves providing your audience with the necessary background information to fully understand your visuals. This may be needed in both an in person or dashboard only presentation.

Business Segment Focus

- Is it clear what part of the business we're looking at?

Comparable Insights

- How good is good?
- Can we compare to industry benchmarks, estimates, or competitor data to establish what good is?

Data Integrity & Ethics

Data Integrity

- Refers to the accuracy, consistency, and reliability of data throughout its lifecycle.
- Crucial to prevent alterations or tampering that can lead to inaccurate insights and harmful decisions.
- Involves verifying sources, ensuring accuracy, and being transparent about data modifications.



Data Integrity & Ethics

Ethics

- Govern responsible behavior in data visualization.
- Important to avoid misrepresentation of data, whether by altering graphical scales or omitting unfavorable data.
- Upholding honesty, acknowledging limitations, avoiding biases, and transparency in methodologies.



Choosing the Right Visual

Table Visual

Tables are the most basic way to present our data, often revealing a huge amount of detail behind each number across a grid.

Share Price Sensitivity						
		Revenue Growth				
Current Price	34.85	0%	5.0%	10.0%	15.0%	20.0%
Exit Multiple	6.0x	19.23	23.73	29.09	35.43	42.89
	7.0x	20.85	25.92	31.97	39.14	47.57
	8.0x	22.48	28.12	34.85	42.84	52.25
	9.0x	24.11	30.31	37.73	46.54	56.94
	10.0x	25.73	32.51	40.61	50.25	61.62

Strengths:

- Straightforward presentation of raw data.
- Suitable for a knowledgeable audience interested in a breakdown of many metrics.
- Particularly useful as reference material or a cheat sheet in fast moving meetings or discussions.

Weaknesses:

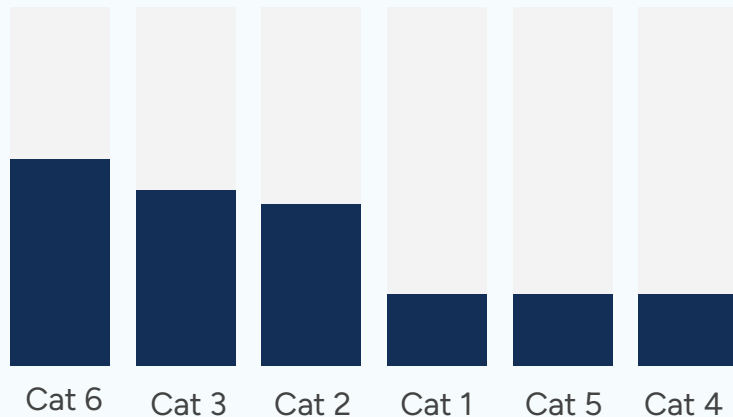
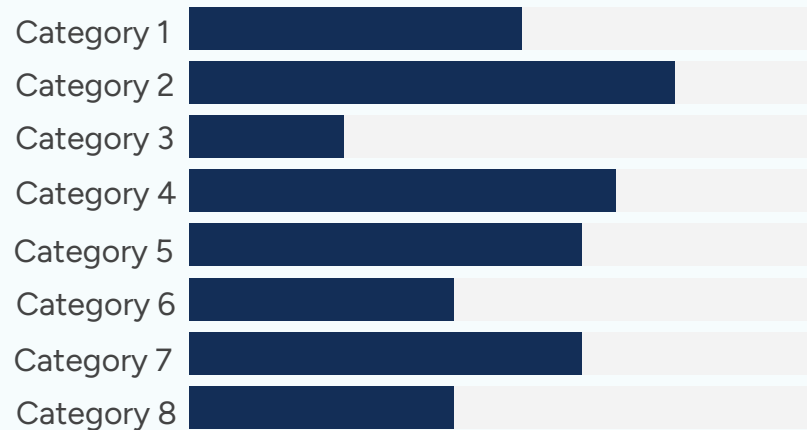
- Often overwhelming with large amounts of data
- Risks losing focus on a key message

When to Use:

- When many metrics are required and a visual won't work.
- When metrics are required as reference material.

Charts for Comparison – Bar / Column

Bars and columns allows us to easily compare the size of different data points or categories.



Strengths:

- Often the best and simplest way to present data.
- Require minimal technical effort to build.

Weaknesses:

- Mostly helpful for a single metric.

When to Use:

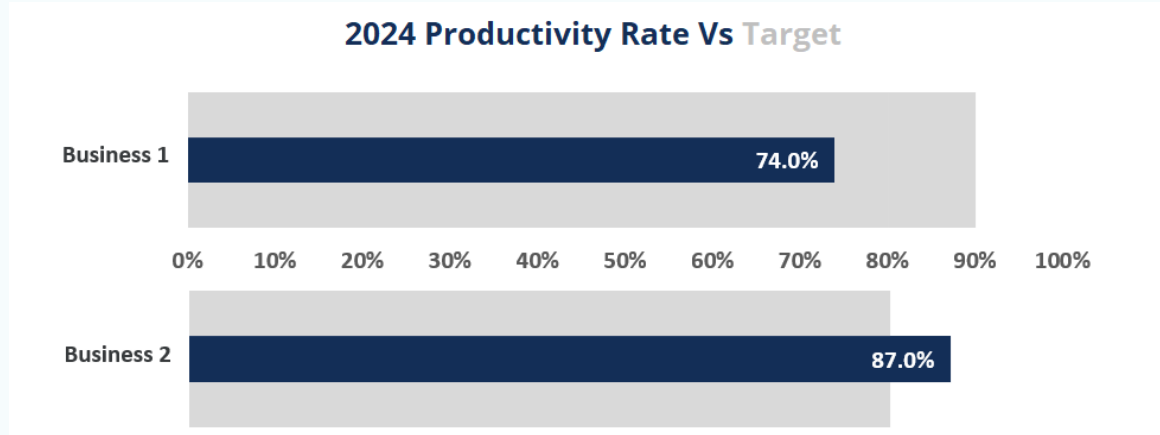
- If a bar or column tells the story well, use it.

Bars Vs Columns

- Columns are often preferred to signify a progression through time (left to right)
- Bars are helpful when category names are long.

Charts for Comparison – Bullet

Bullet charts are a variation on a bar or column chart and include a benchmark metric for comparison.



Strengths:

- Ideal for comparing separate categories to independent targets or benchmarks.
- Offers a quick snapshot of performance relative to expectations.

Weaknesses:

- Must be made explicitly clear which bar is the target.

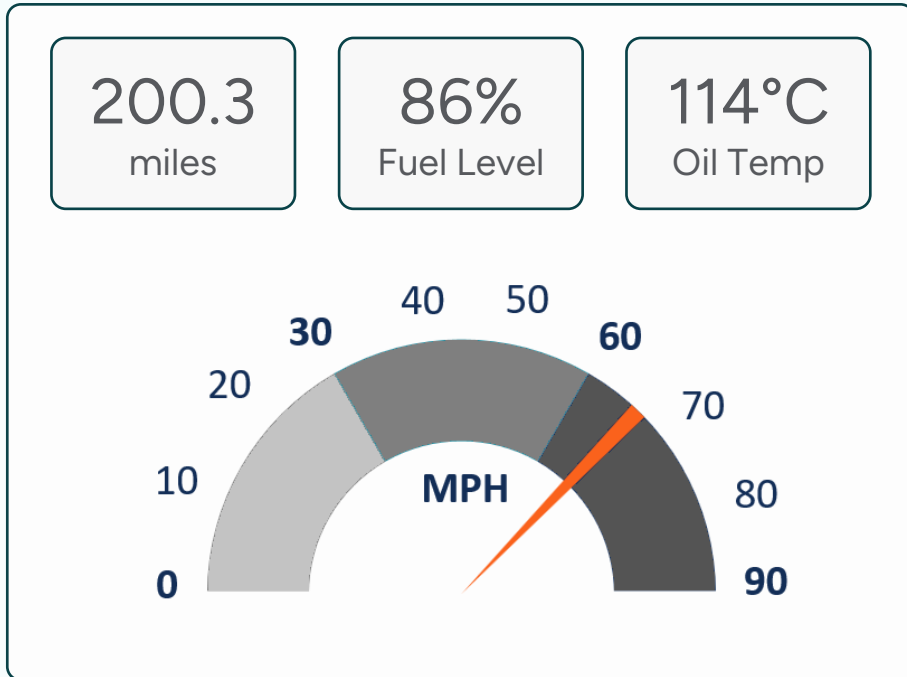
When to Use:

- When each category has an independent target or benchmark.

“Bar and column charts are often the best and simplest way to get your message across.”

Charts for Highlighting Key Numbers

Card and Gauge visuals put critical and recognisable numbers front and centre, with minimal distractions.



Strengths:

- Direct and concise presentation of key information.
- Clear messaging reduces the chance of misunderstanding.

Weaknesses:

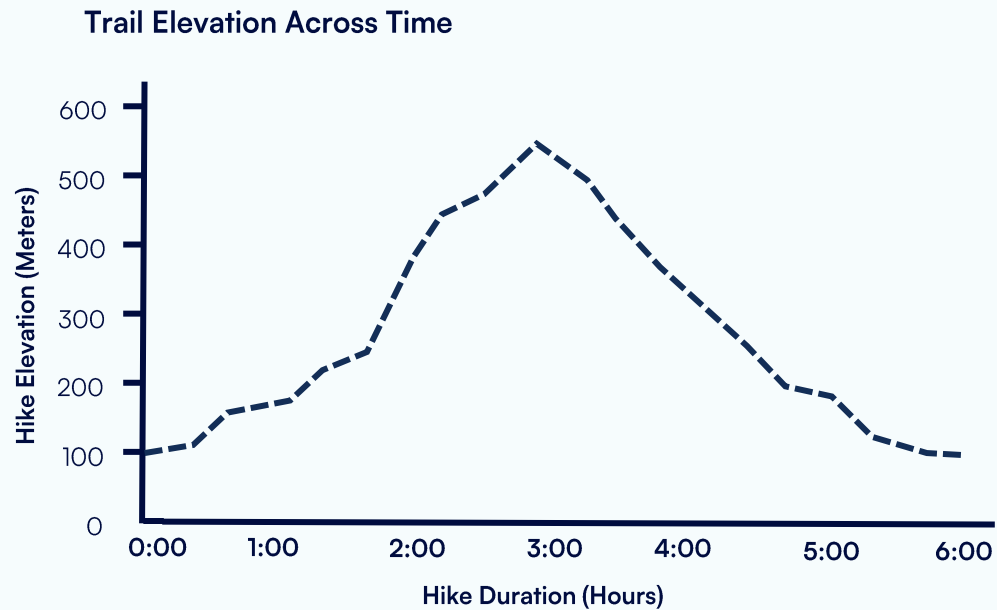
- Limited narrative or context.

When to Use:

- To provide an at-a-glance check for a recognizable and important metric.

Charts for Trends

A line or area chart can help us see how a metric changes over time.



Strengths:

- Ideal for depicting changes over time
- Clear visualization of trends, like stock prices or yearly temperatures

Weaknesses:

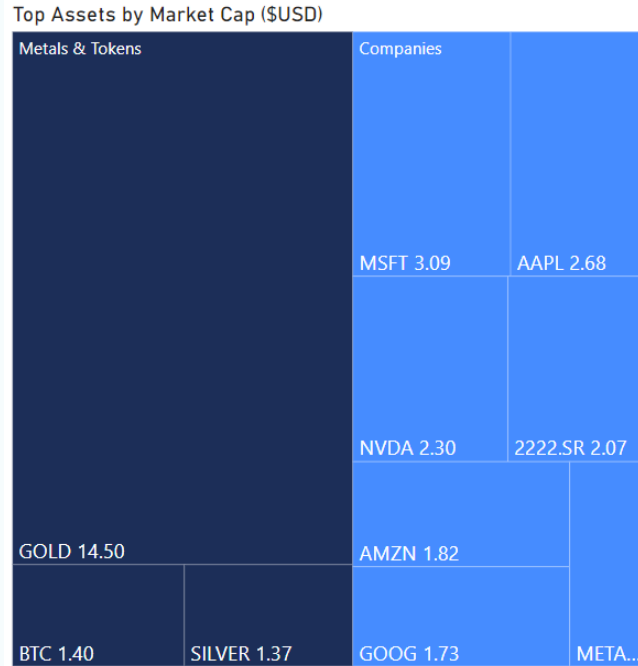
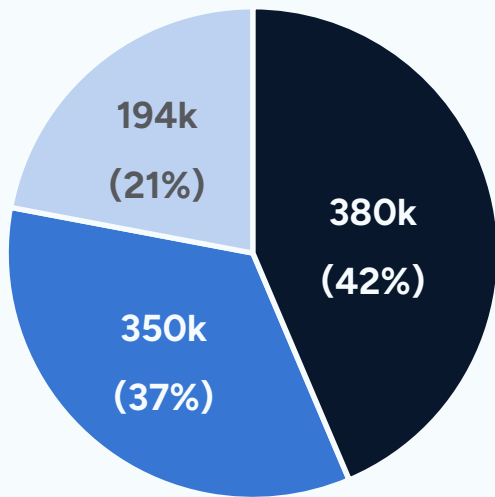
- Can be misleading without proper context or scales

When to Use:

- When you want to show how something changes over time

Charts for Contribution – Pie & Treemap

Pie charts and tree maps help us see which categories are most important or which have the most of something.



Strengths:

- Helpful to give a quick sense of importance or relative contribution.

Weaknesses:

- Difficult to interpret accurately due to challenge in comparing slice sizes (angles) or rectangle sizes.

When to Use:

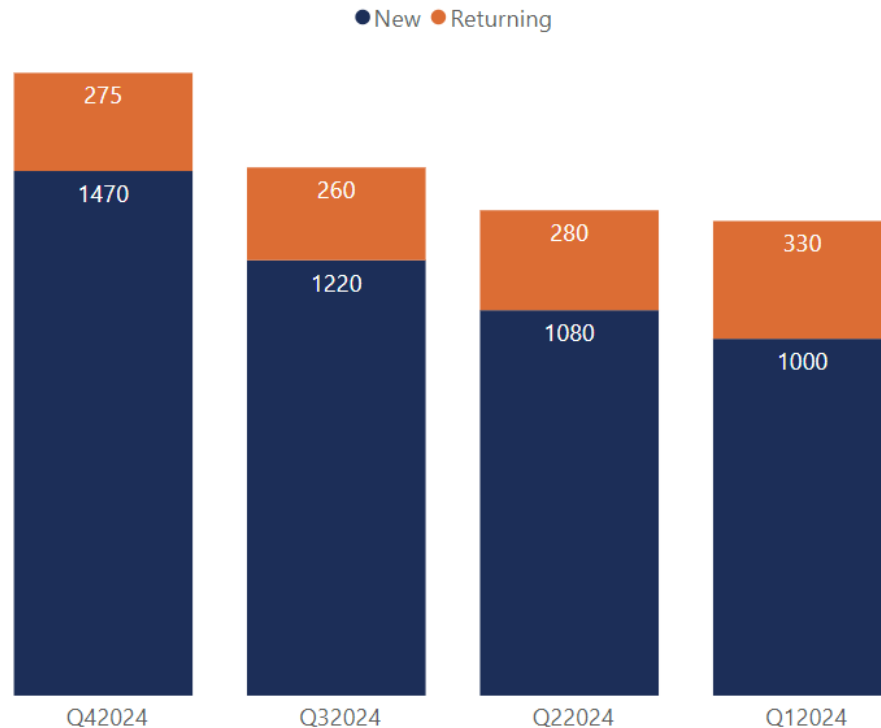
- Only use pie charts with few categories. For more categories, use a bar/column chart.

Pie Chart Top Tip: Always start the largest segment at 12 o'clock, and move clockwise in descending order.

Charts for Both Comparison & Contribution – Stacked Bars

Stacked bar charts help us see the contribution to a bigger picture over time.

Customers by Type, per Quarter



Strengths:

- Excellent for displaying contributions to the total in each column
- Enables a basic comparison over time.

Weaknesses:

- Can become messy with too many subcategories
- Generally, allows for effective visual comparison of only the most important category.

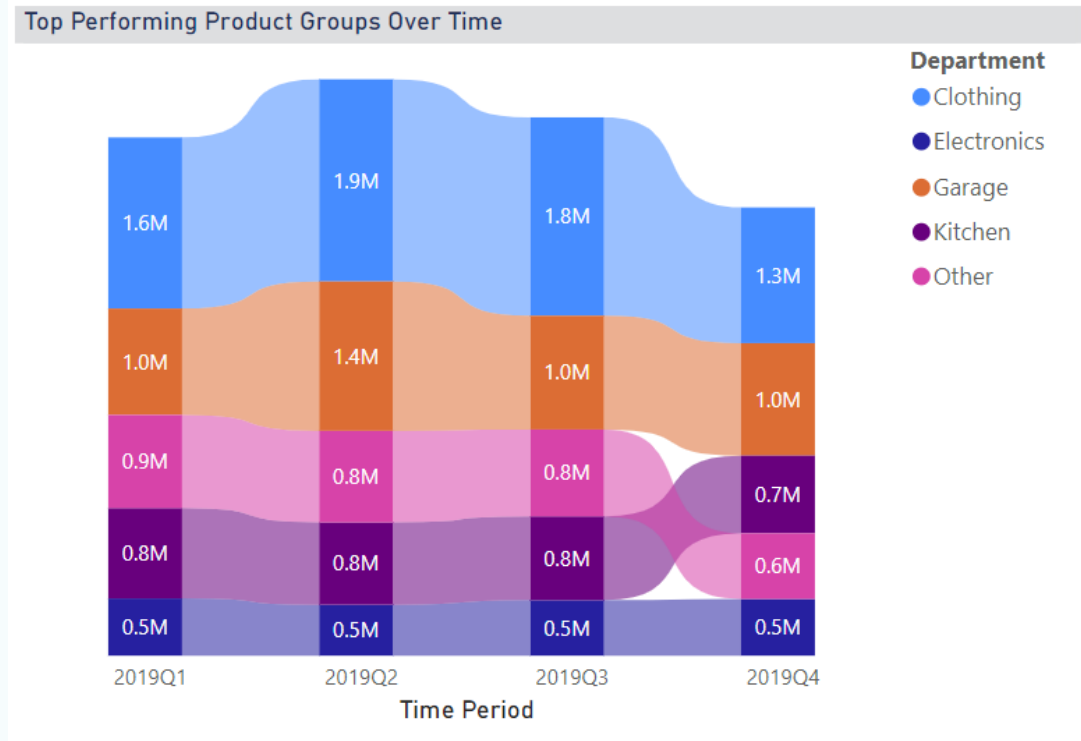
When to Use:

- Use when the contribution of 1 or 2 categories over time is particularly important.

Stacked Chart Top Tip: Labels help to make comparisons across non-primary categories.

Charts for Both Comparison & Contribution – Ribbon

Ribbon charts allow the order of stacked columns to change, showing visually the change of rank.



Strengths:

- Helpful if showing change across time is the main goal
- Helps us visualize time from left to right.

Weaknesses:

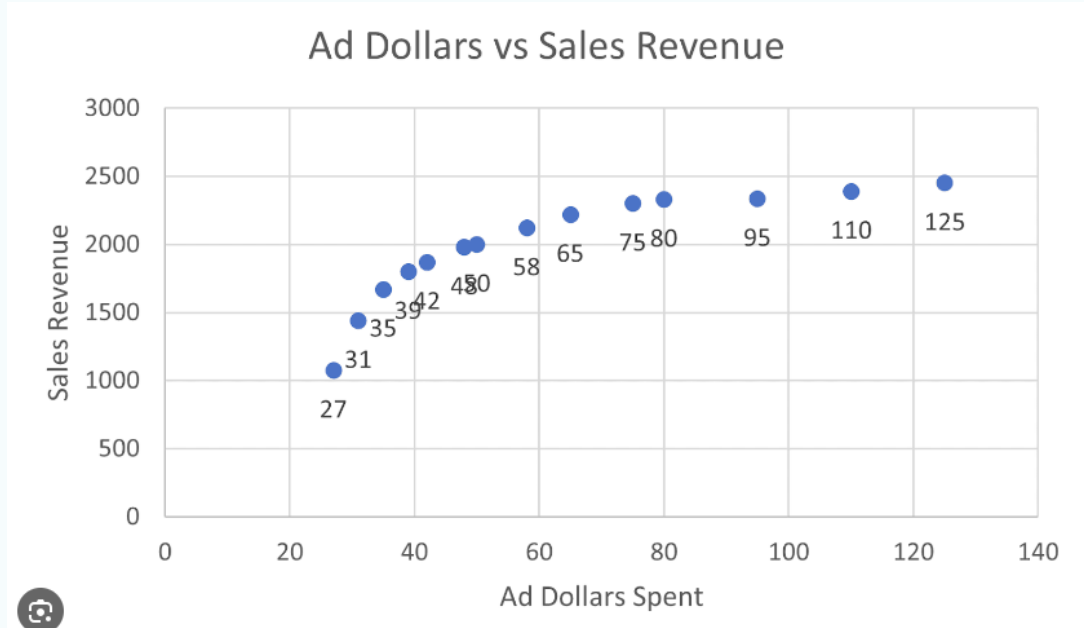
- Can become confusing if data shows a lot of volatility

When to Use Ribbon Charts:

- Use with caution, recognizing the complexity they introduce

Charts for Relationships – Scatter

Scatter charts allow us to plot two continuous metrics against each other.



Strengths

- Helps understand the relationship between two numeric variables

Weaknesses

- Correlation doesn't imply causation. Audiences must be cautious.
- Strength of relationship is not measured precisely

When to Use:

- When you have two numeric variables.

Scatter Top Tip: Too many grid lines are messy. Too few and comparison of points becomes difficult. Make gridlines light in color.

Charts for Relationships – Correlation Matrix

A correlation matrix shows the correlation coefficients between pairs of variables.

	Sales Revenue	Advertising Spend	Customer Satisfaction Scores	Number of Returned Items
Sales Revenue	-	0.85	0.20	-0.30
Advertising Spend	0.85	-	0.25	0.10
Customer Satisfaction Scores	0.20	0.25	-	-0.45
Number of Returned Items	-0.30	0.10	-0.45	-

Strength:

- Help us understand the strength of relationships between numeric variables.

Weaknesses:

- Risk of information overload for an unfamiliar or non-technical audience.

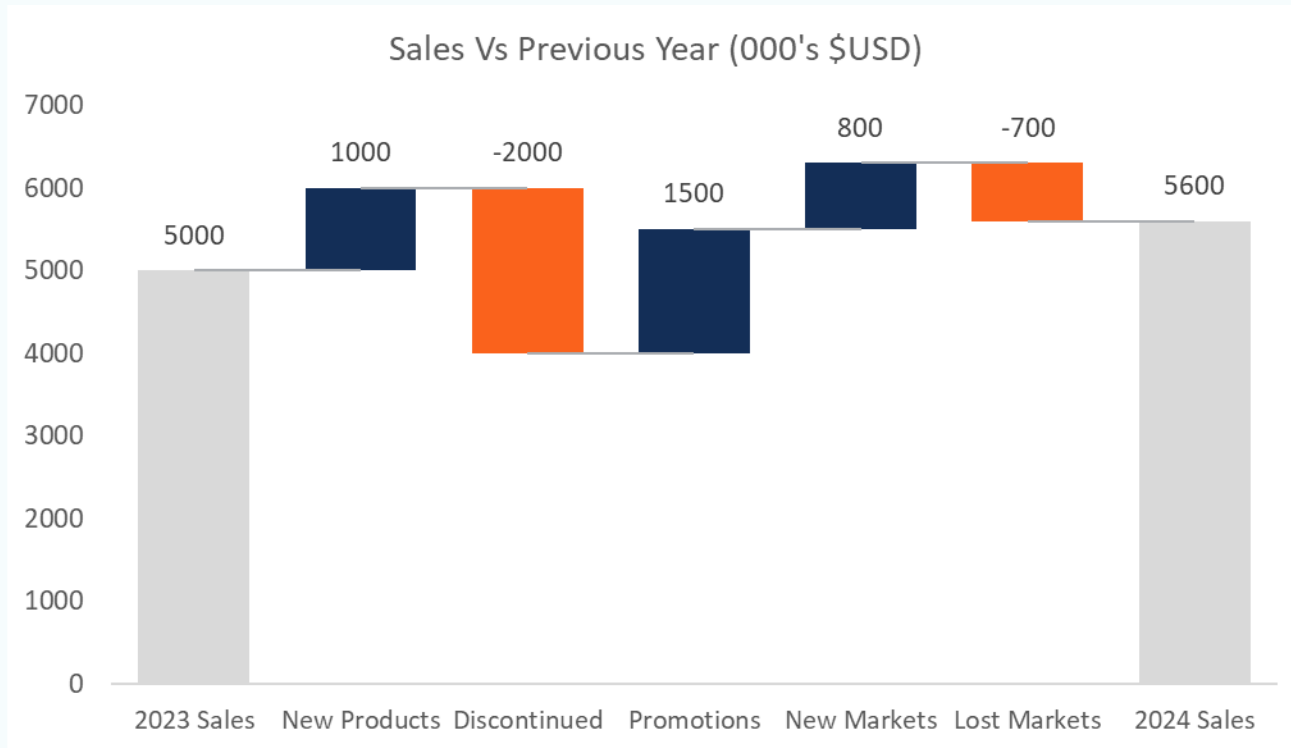
When to Use:

- When you have a number of different numeric variables and want to understand the strength of the relationship between each pair.
- Particularly helpful in data science projects like regression.

Correlation Matrix Top Tip: Formatting is key to creating a great correlation matrix. More in the next chapter on use of color.

Other Charts – Waterfalls

Waterfall charts visualize the impact of sequential factors, showing progression from start to finish.



Strengths:

- Provides a clear narrative from start to finish
- Commonly used in finance to explain variations in sales year-over-year or differences between target and actual values.

Weaknesses:

- Can become confusing without clear labelling, especially with many intermediate steps

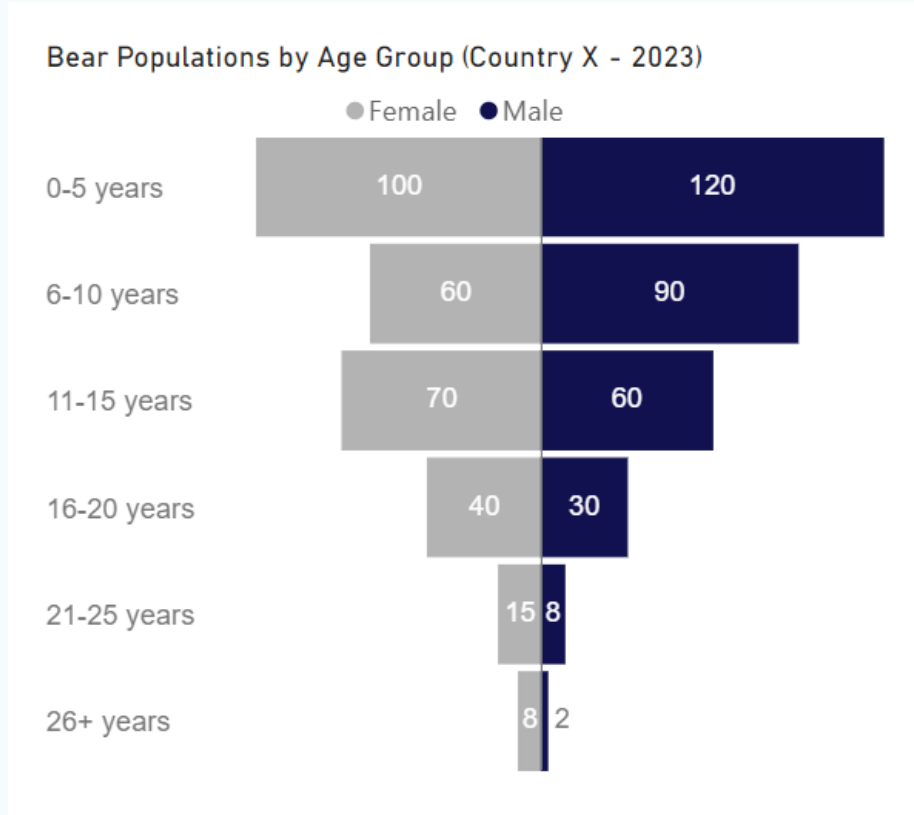
When to Use:

- Use to show progression and show how different factors contribute to a final result

Waterfalls Top Tip: The lines joining each bar to the next are key to understanding the continuity of the chart.

Other Charts – Tornado

Tornado charts help us visualize a metric between two segments and across and categories.



Strengths:

- Ideal for comparison across ordered categories or showing a single metric at different stages of a process.
- Provides a straightforward way to compare two directions simultaneously.

Weaknesses:

- Can be hard to compare absolute number from left to right.

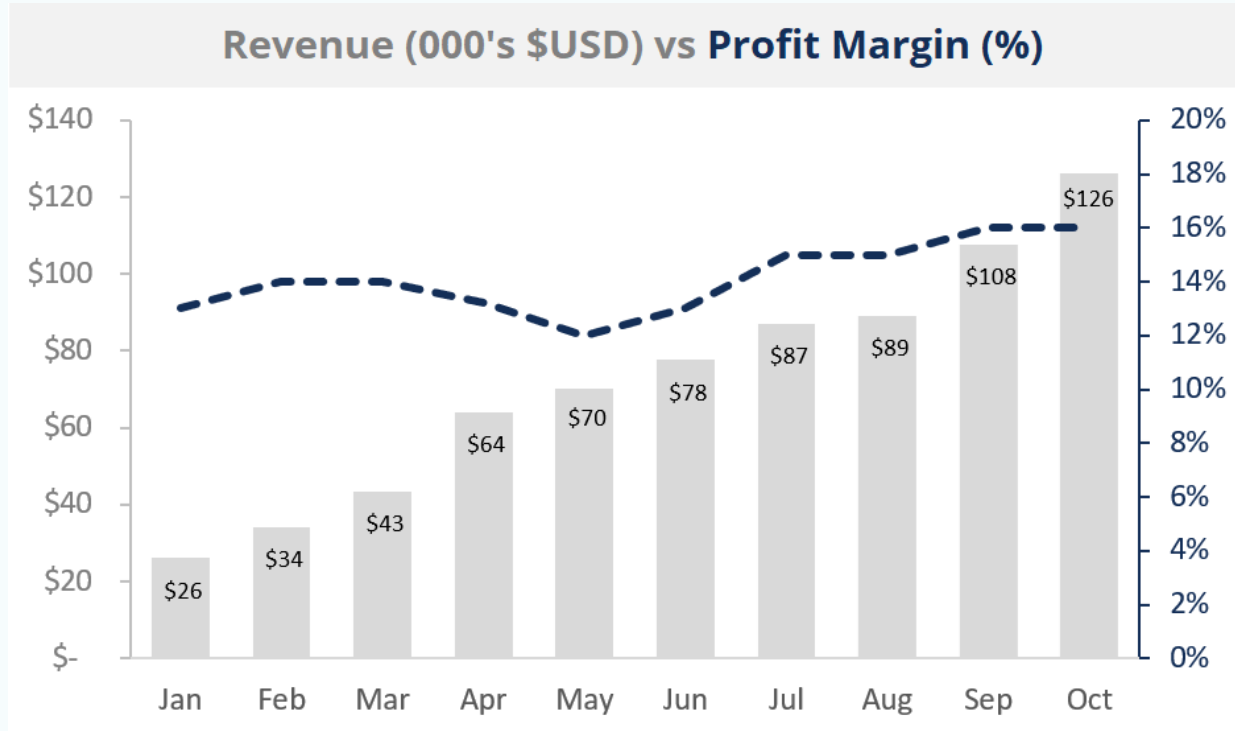
When to Use:

- When there are 2 clear groups that can be compared across a number of categories.

Tornado Top Tip: Labels help to make comparisons between left and right.

Combination Charts

Combination charts allow us to combine two visuals into one in order to compare two variables.



Strengths:

- Ideal when showing two different types of units, e.g., dollars and percentages, or simply two different variables.

Weaknesses:

- Added complexity, requiring more effort to interpret compared to single-type charts.

When to Use:

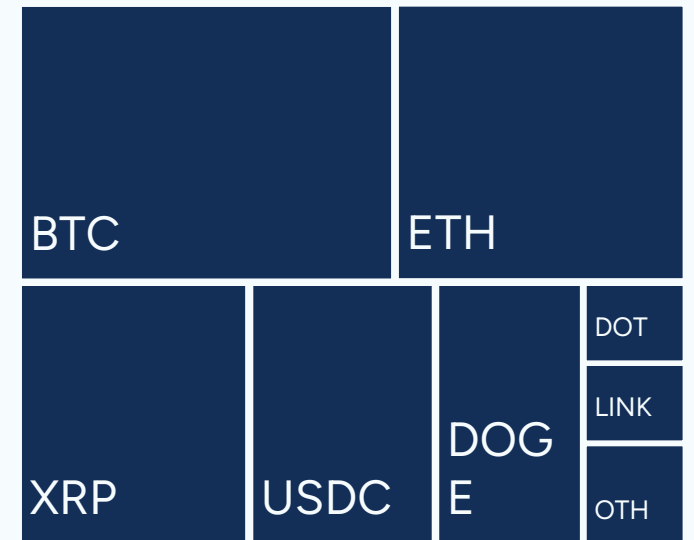
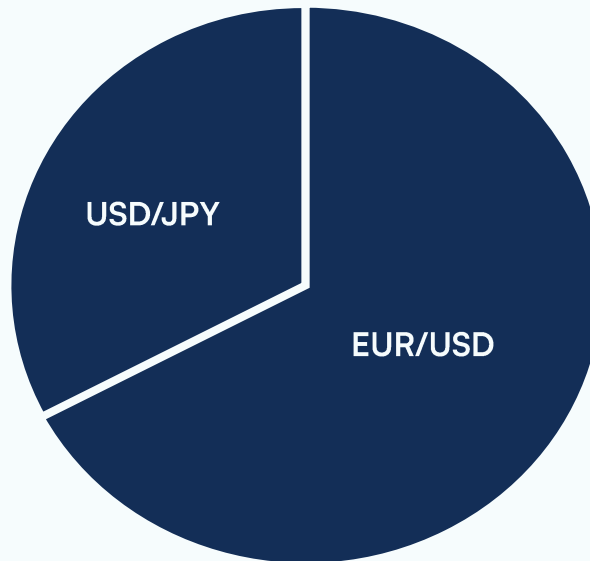
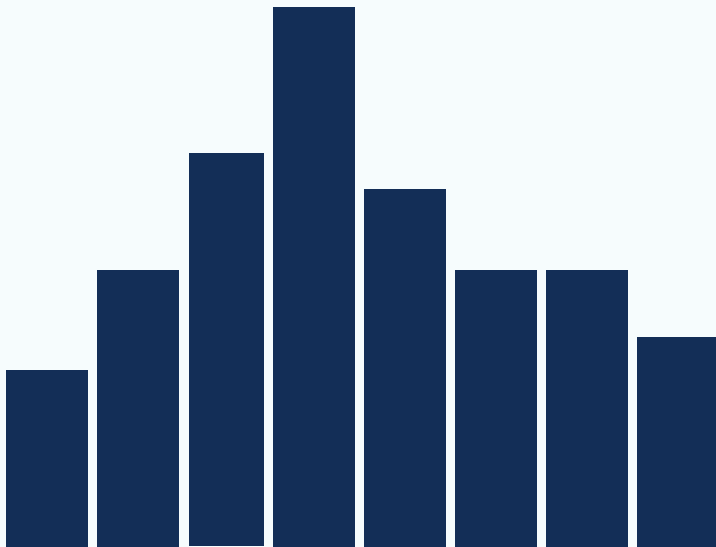
- Suitable when a single chart type is insufficient for providing necessary context or when multiple data layers need to be compared.

Combination Charts Top Tip: It is important to clearly indicate which axis corresponds to each data series, ideally with use of color.

Design & Focusing Attention

Size, Position & Order

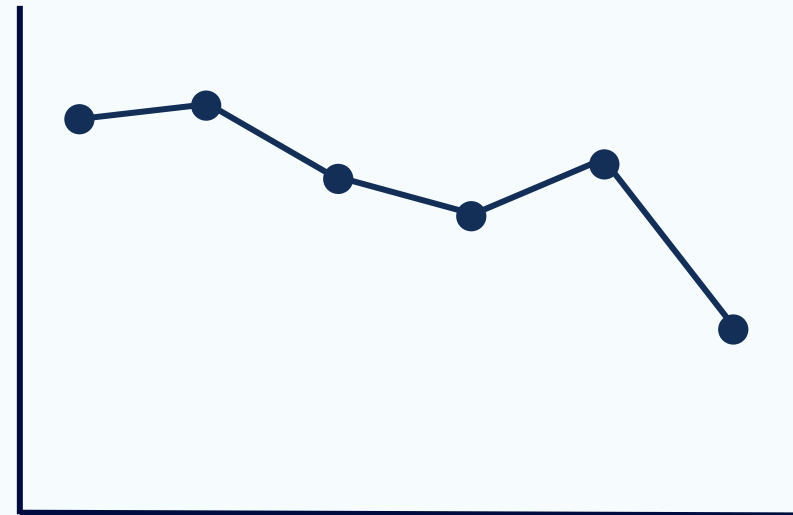
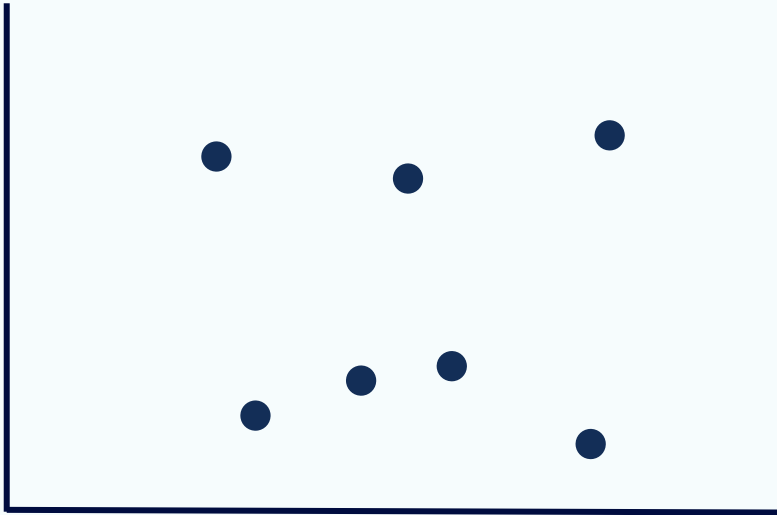
Size is a common visual attribute that we can use to communicate absolute or relative performance.



Size enables us to quickly compare data points.

Size, Position & Order

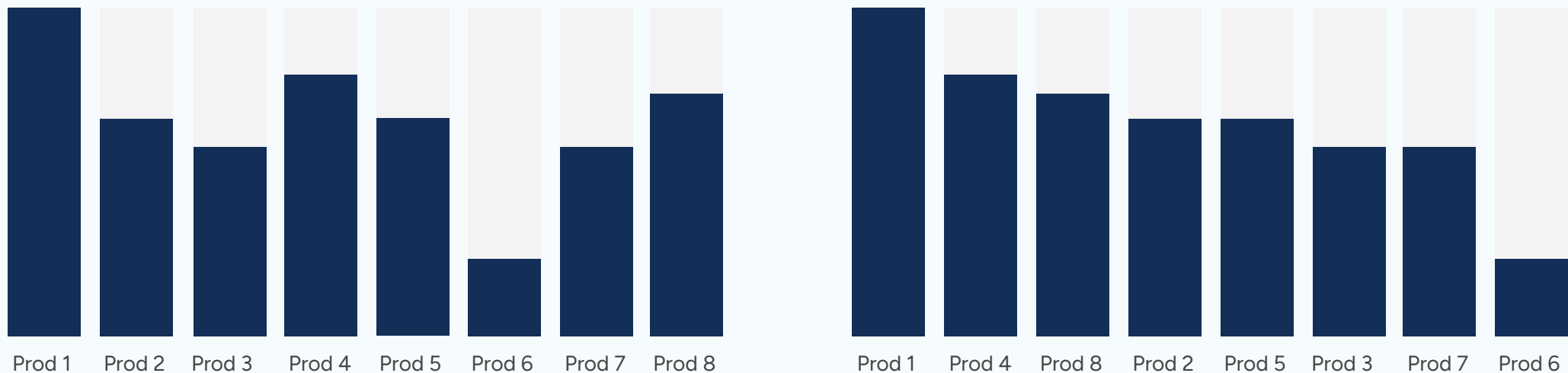
Position helps us immediately evaluate which values are higher and identify trends.



We can deduce all of this without reading any numbers.

Size, Position & Order

Order allows us to draw conclusions about the relative size and importance of categories.



All of our changes aim to reduce the thought required to extract the key message.

How We Interpret Color



279,427,332



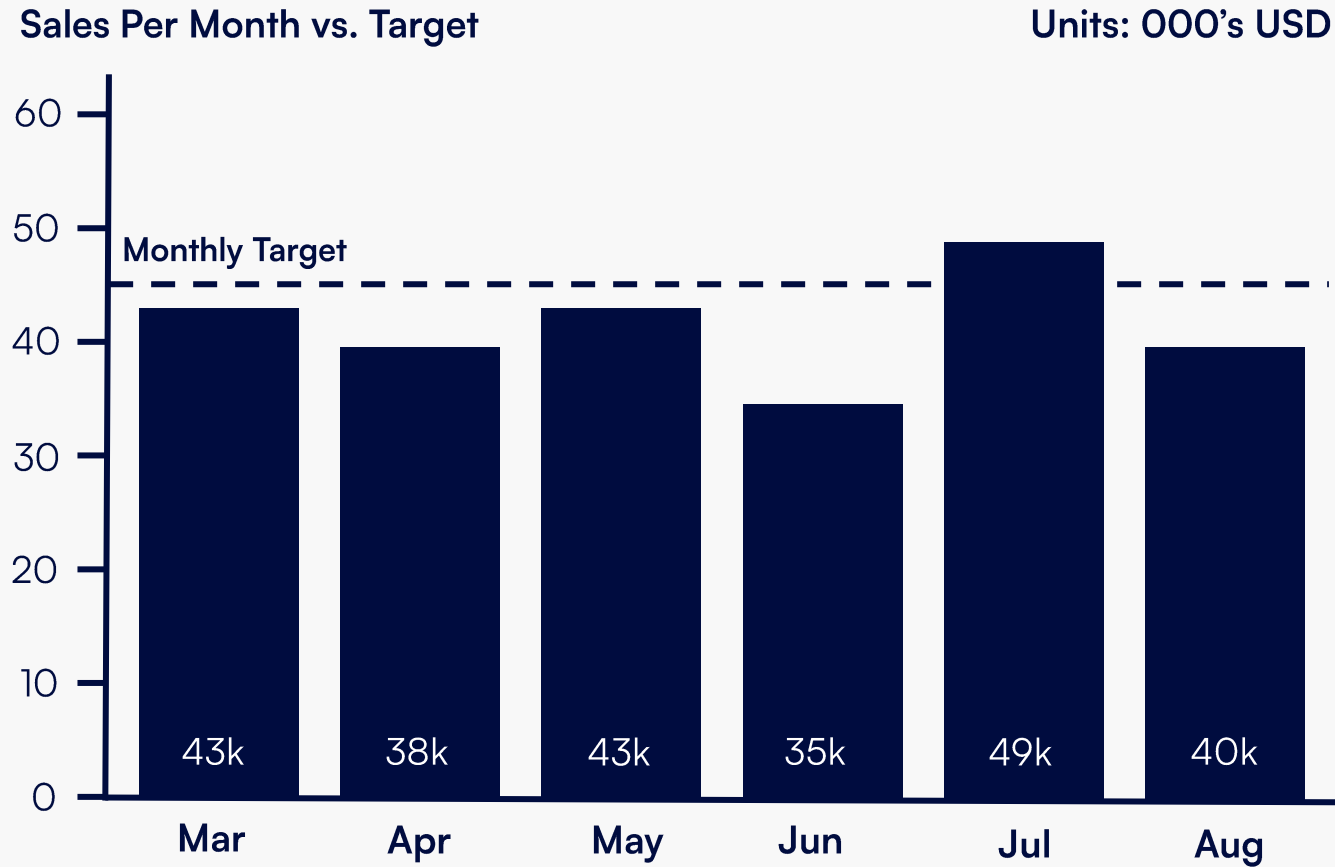
279,422,332

How We Interpret Color

Header	Amount 1	Amount 2	Amount 3
Rent & Utility Cost	276,375,332	276,375,332	136,662,093
Insurance	206,928,200	205,782,025	199,522,452
Car Experience and Fuel	186,837,363	186,414,067	184,418,245
Food	220,932,350	165,931,087	276,601,938
Household Goods	255,082,851	202,990,579	187,950,109
Clothing	231,121,811	242,303,655	164,085,280
Furniture	183,836,865	178,826,761	260,516,793

Numbers take a huge amount of cognitive load to interpret. Color makes value comparisons much faster.

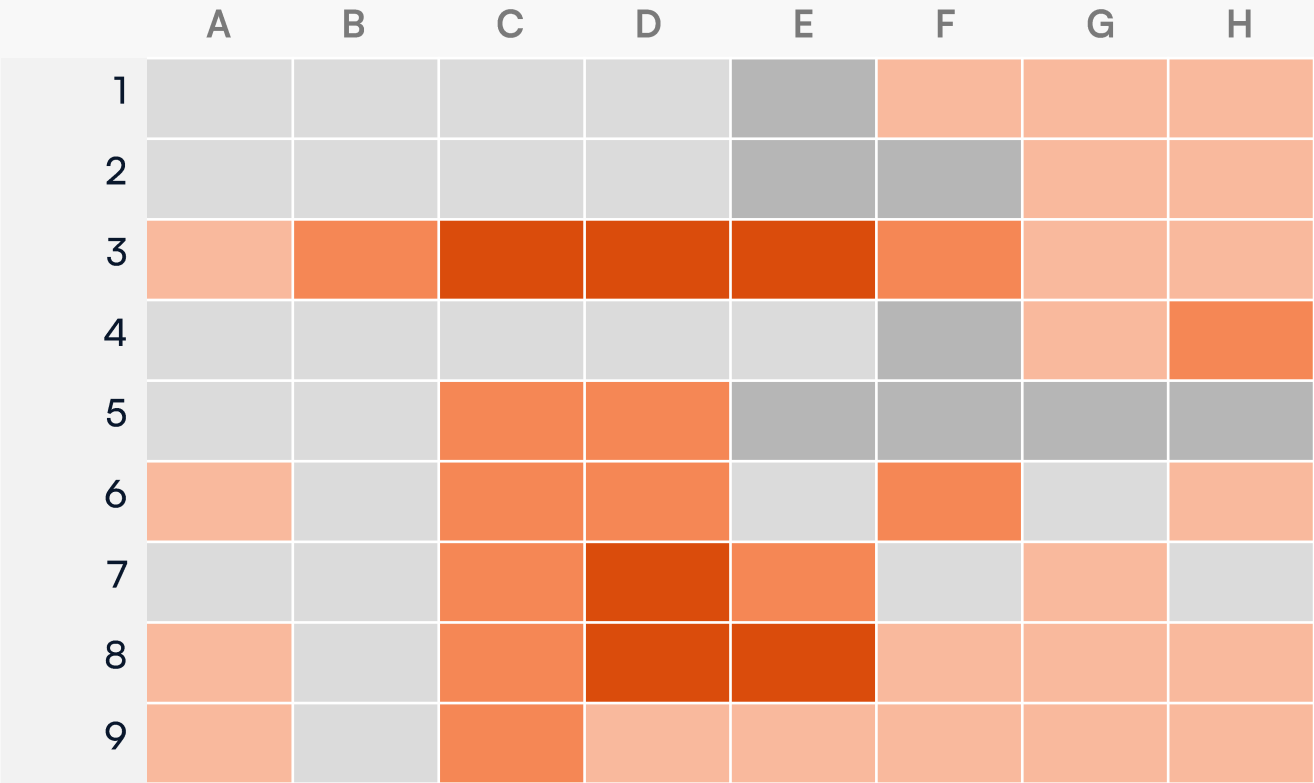
Using Color to Indicate Performance



- **Single colors make it difficult** to focus on a key message.
- **Multiple colors** can help indicate performance but **may be overwhelming**.
- Using **grey for less important months** can help reduce color overload.
- **Lighter grays** remind us those bars are **not the focus**.

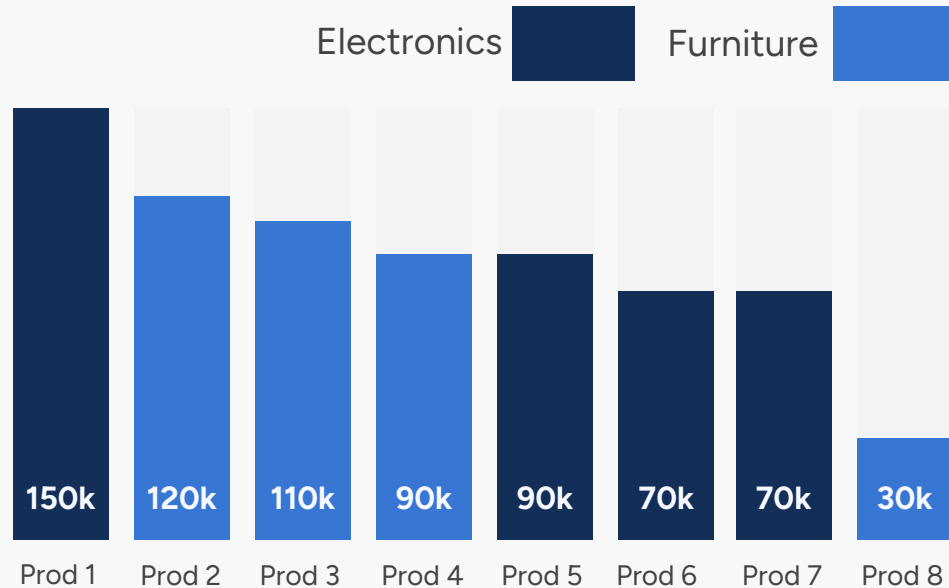
Using Color to Indicate Performance

Identification of Poor Performing Battery Cells



Using Color to Indicate Association

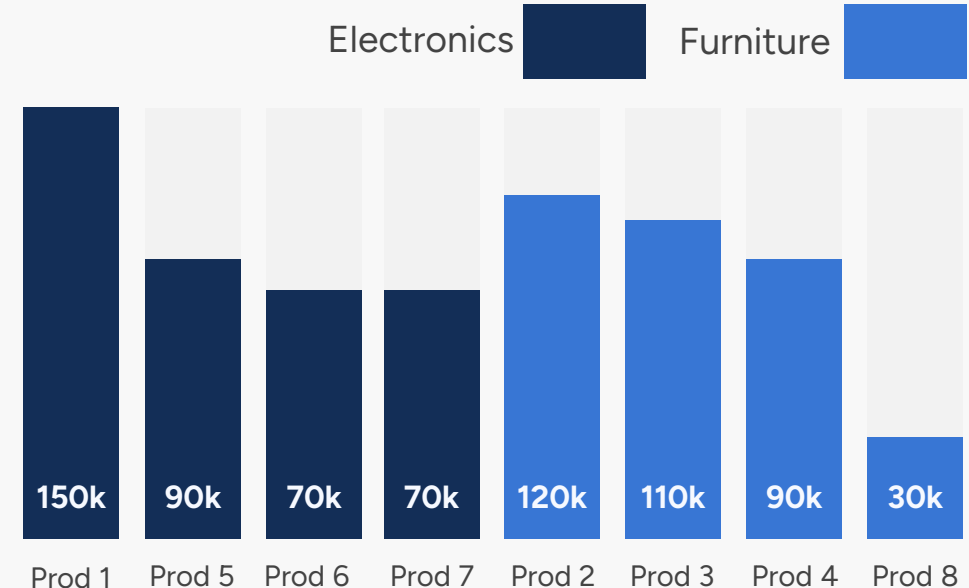
Sales by Product & Category (\$USD)



Colored Column Chart (Ranked)

- Easier to interpret when colors represent categories

Sales by Product & Category (\$USD)

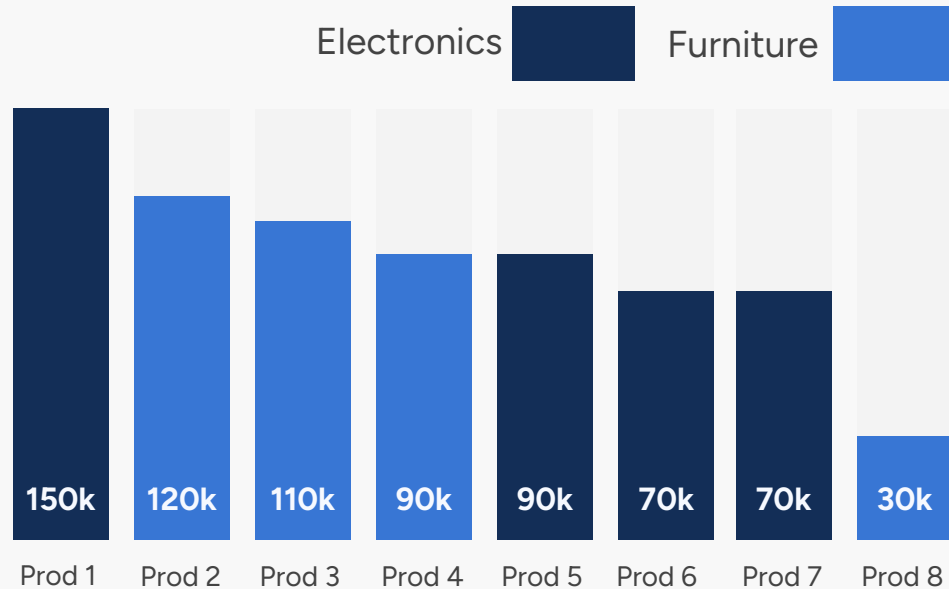


Colored Column Chart (Group Ranked)

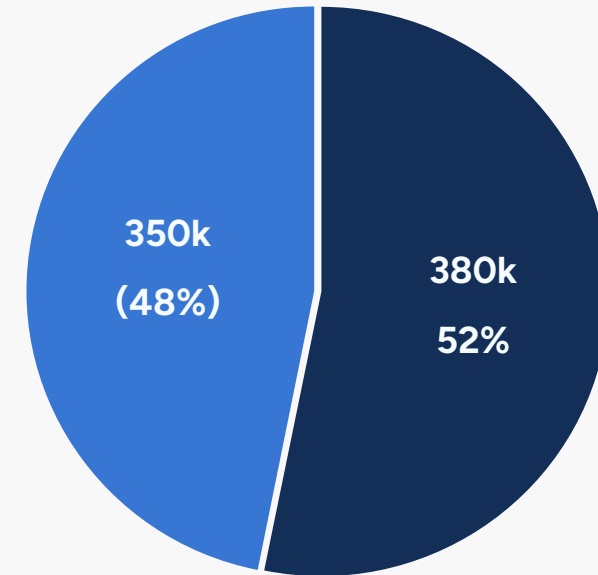
- Grouping columns by category may help us answer certain questions

Using Color to Indicate Association

Sales by Product & Category (\$USD)



Total Sales by Category (\$USD)

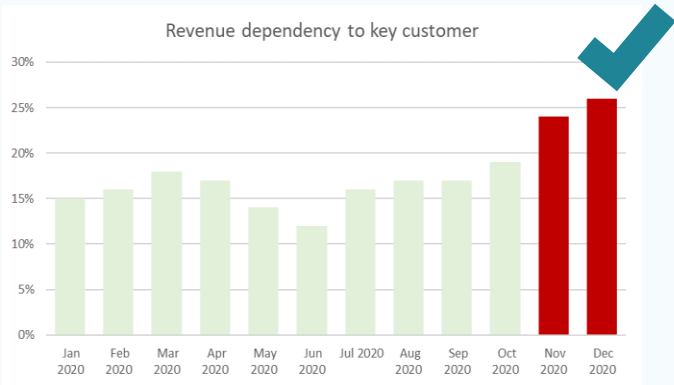
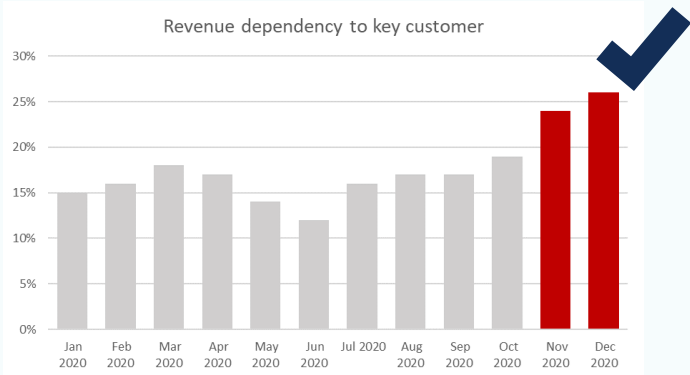
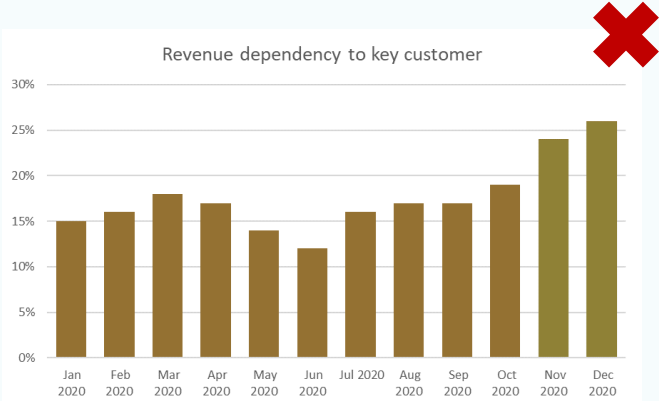
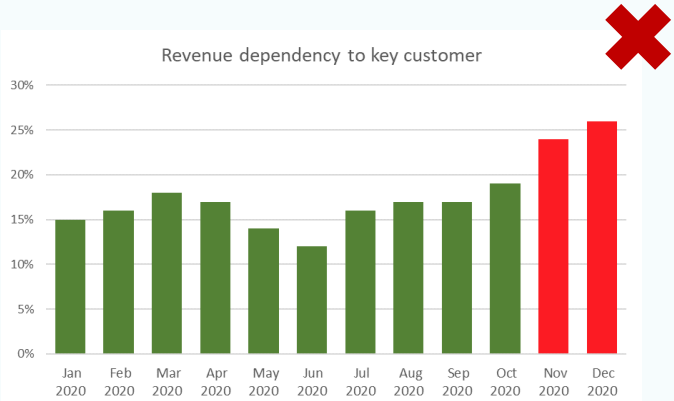


Association Relies on Consistency

- Category colors in our first visual should match those in our second.
- Consistency reduces the amount of processing for the audience, making interpretation faster.
- Stick to generally accepted color or brand associations; for example, Ferrari is always Red.

Color Blindness

 Color blindness affects 8% of men and 0.5% of women.



Red-green charts, in particular, may be difficult to read for a color-blind audience.

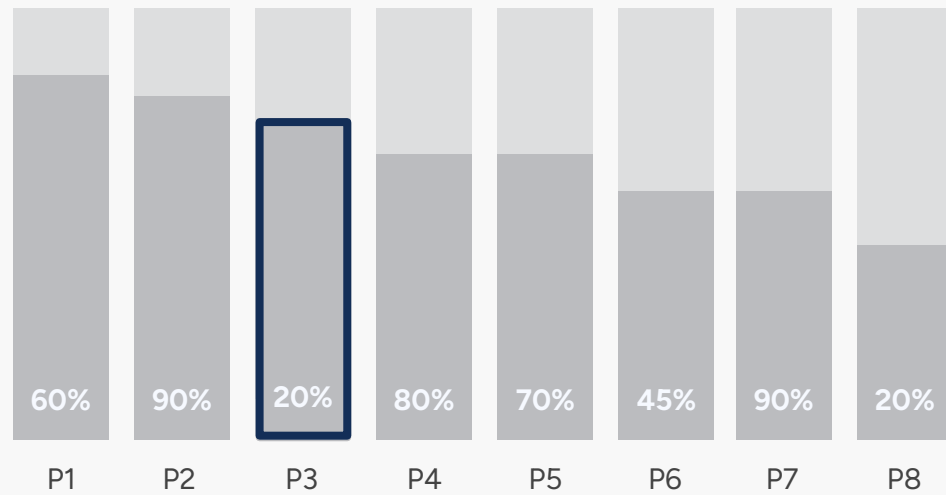


Consider only coloring your key message.



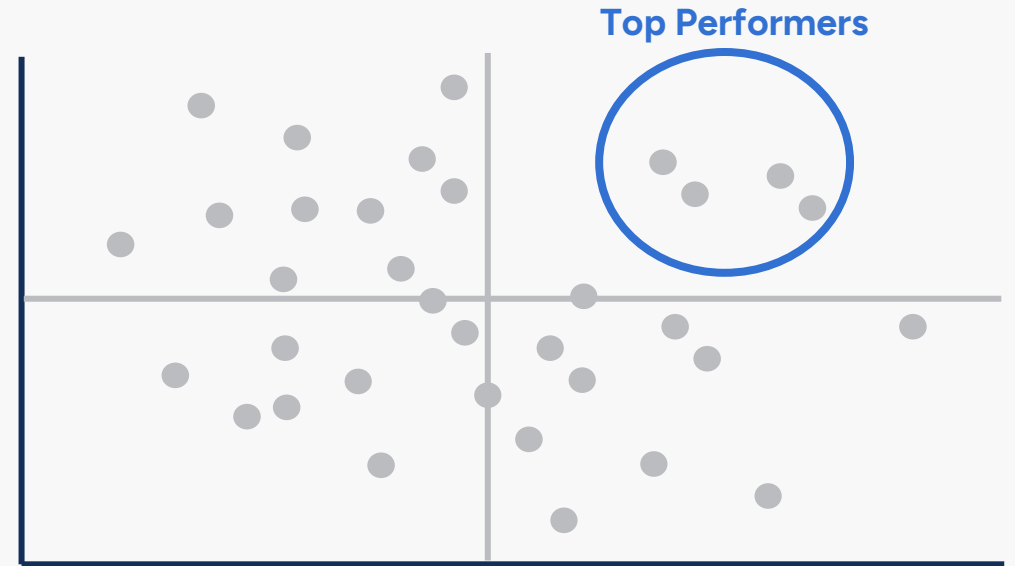
If you must use red/green, consider using high and low brightness or contrast .

Enclosure



Enclosed Column

- Helps to draw attention to a specific data point

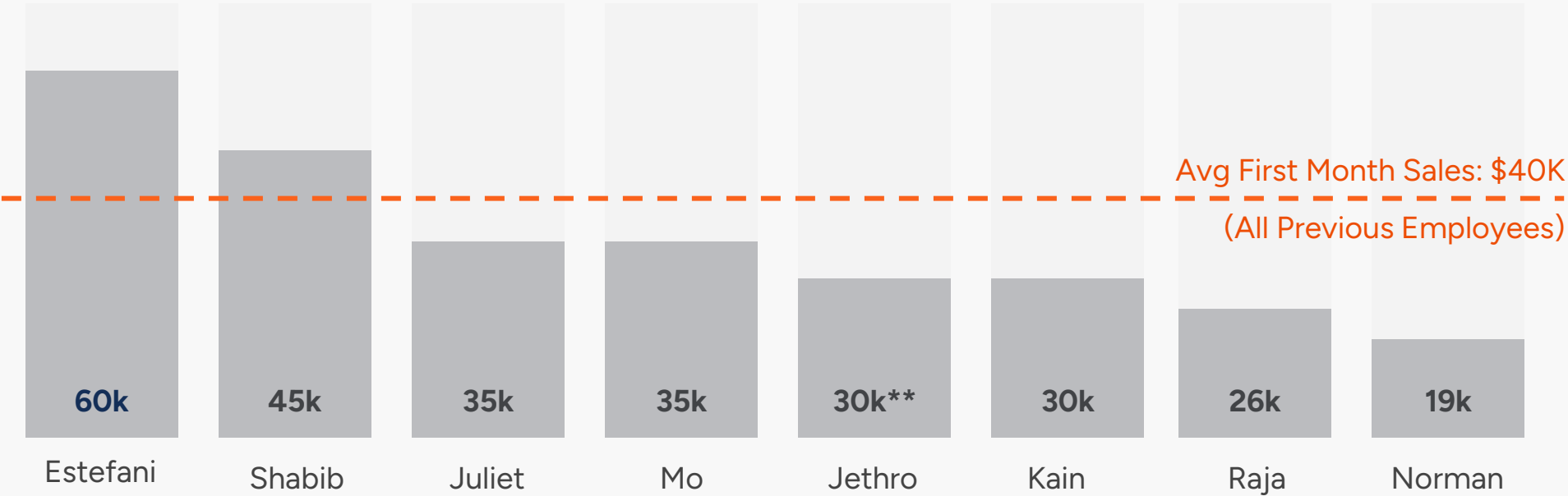


Enclosed Data Points

- Helps to draw attention to a group of data points
- May suggest a similar characteristic or trait
- Annotation can be used to explain the enclosure

Annotations & Context

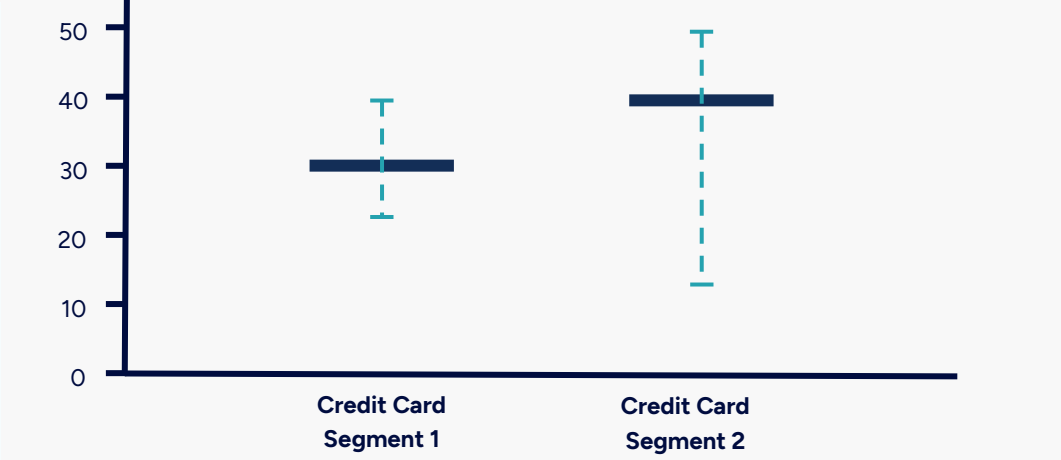
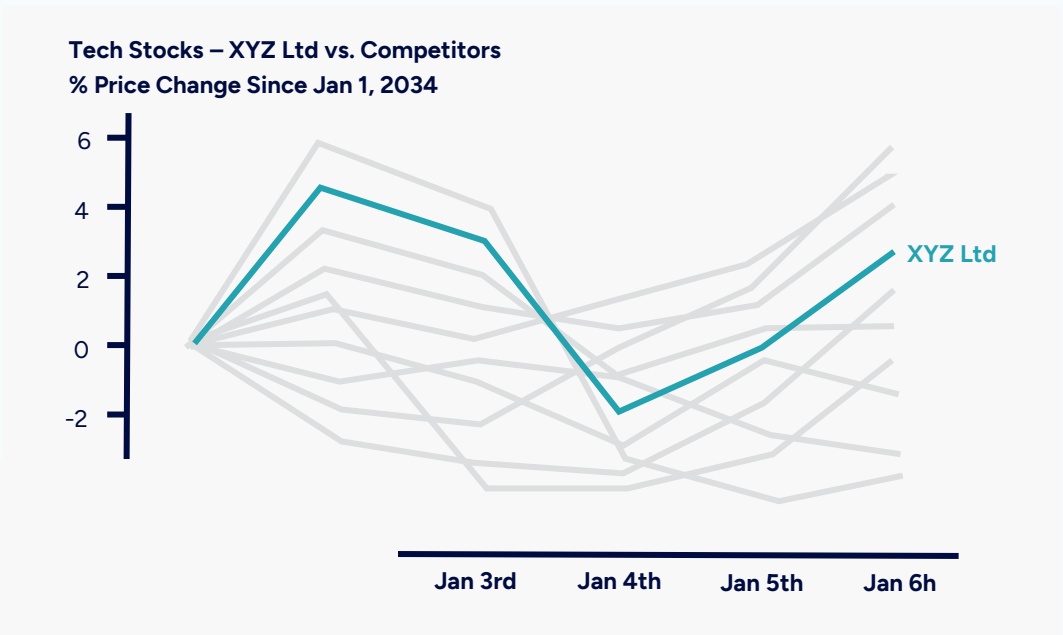
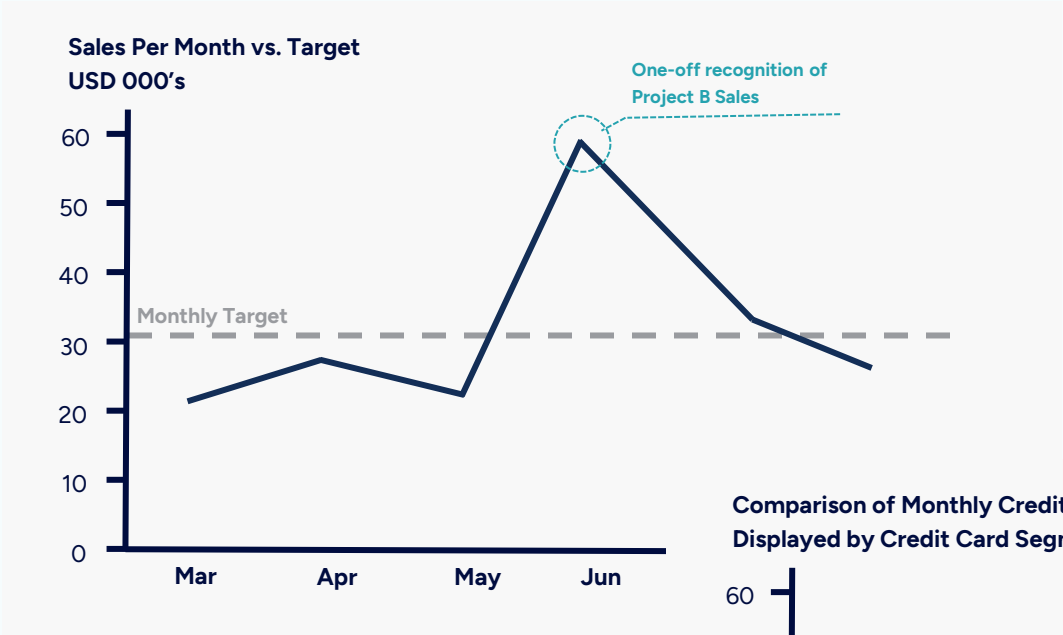
First Month Sales (\$USD) – New Staff Feb 2024



In Summary

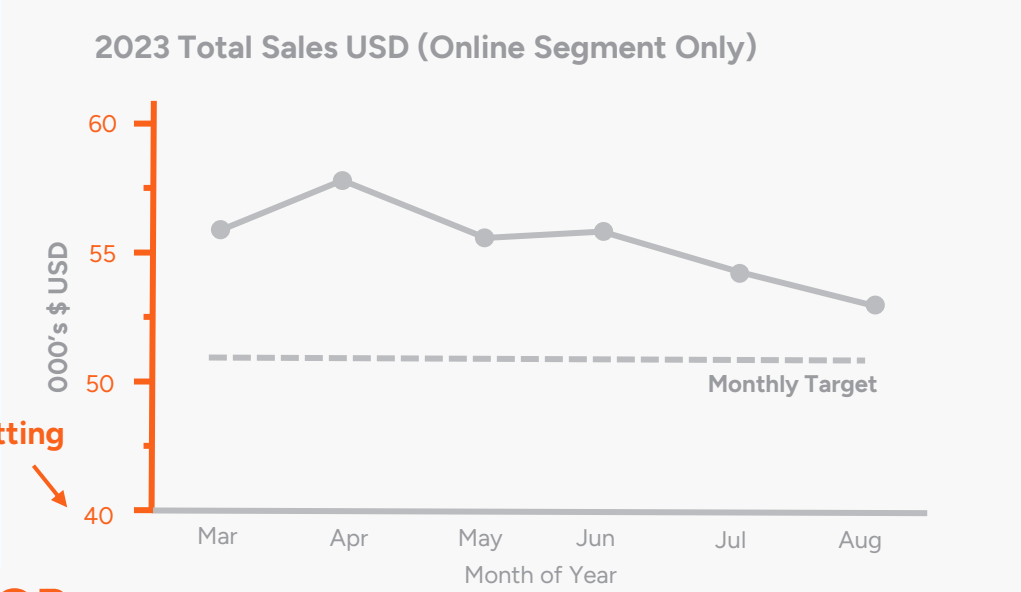
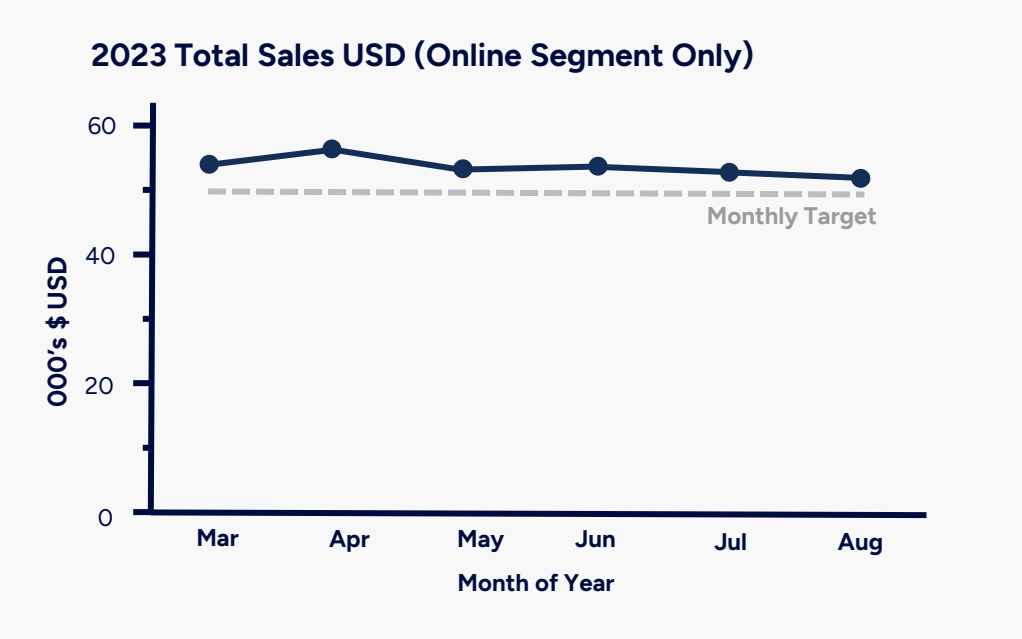
Annotations can completely change the way we interpret key messages.

Annotations & Context

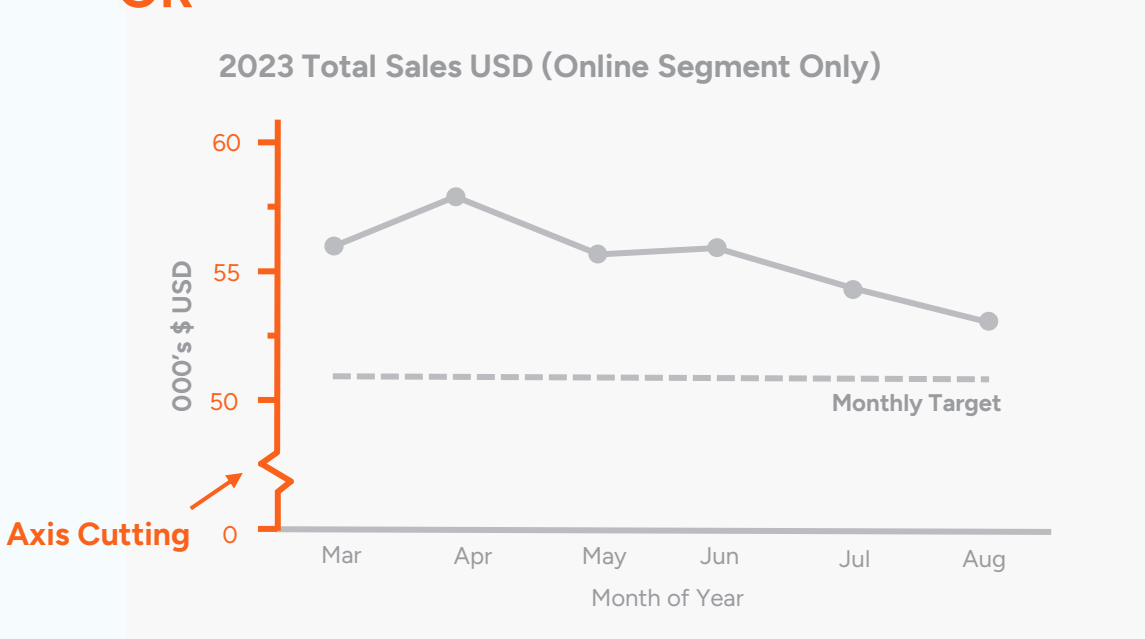


Titles, Units & Axis

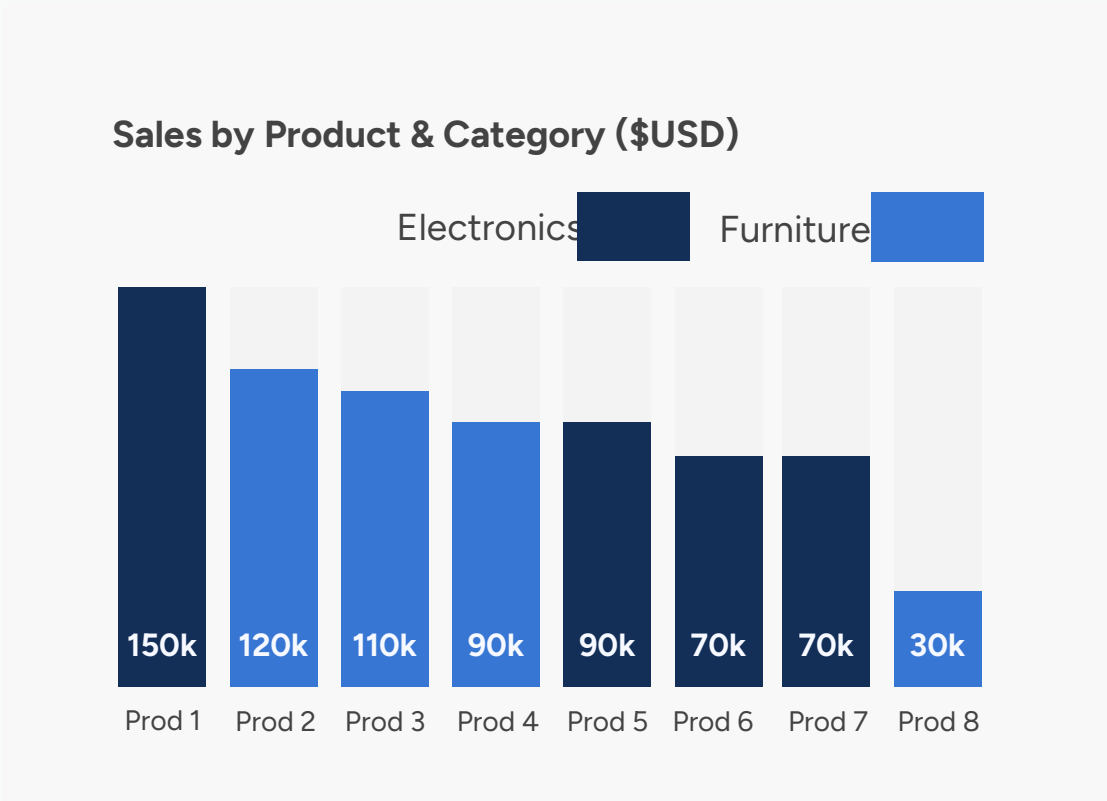
A good chart can be ruined without a clear title, units, or axis labels.



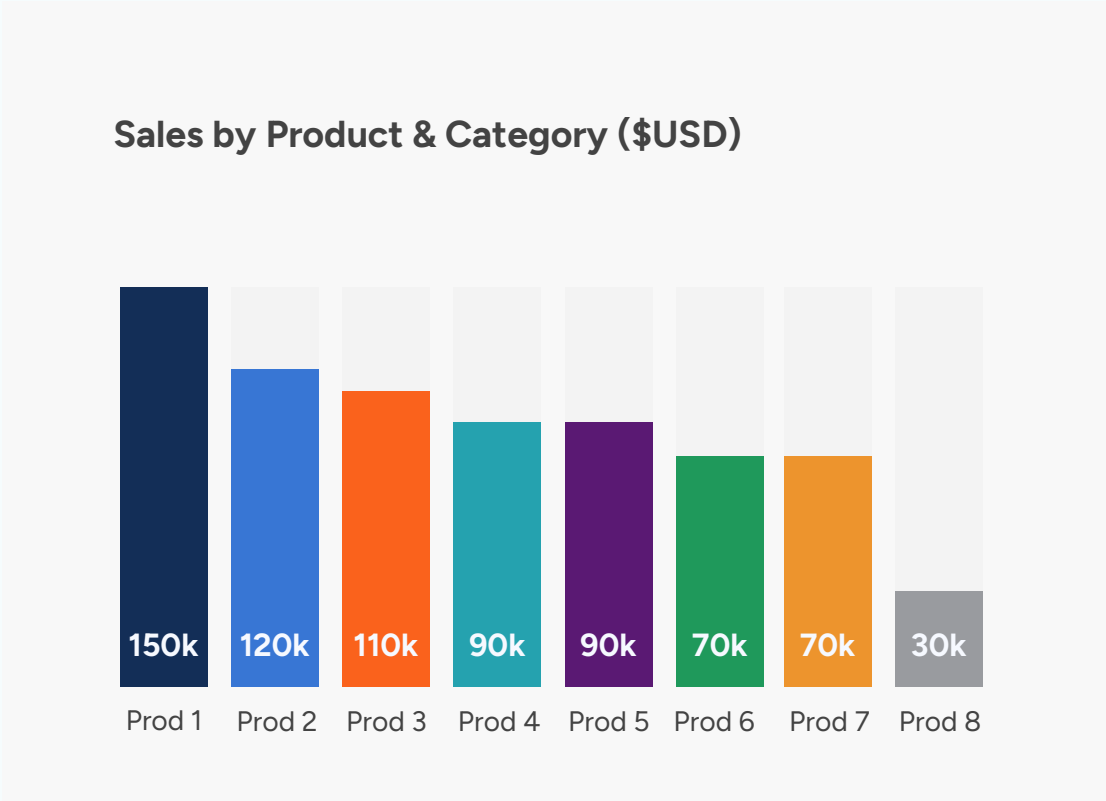
OR



Design & Aesthetics

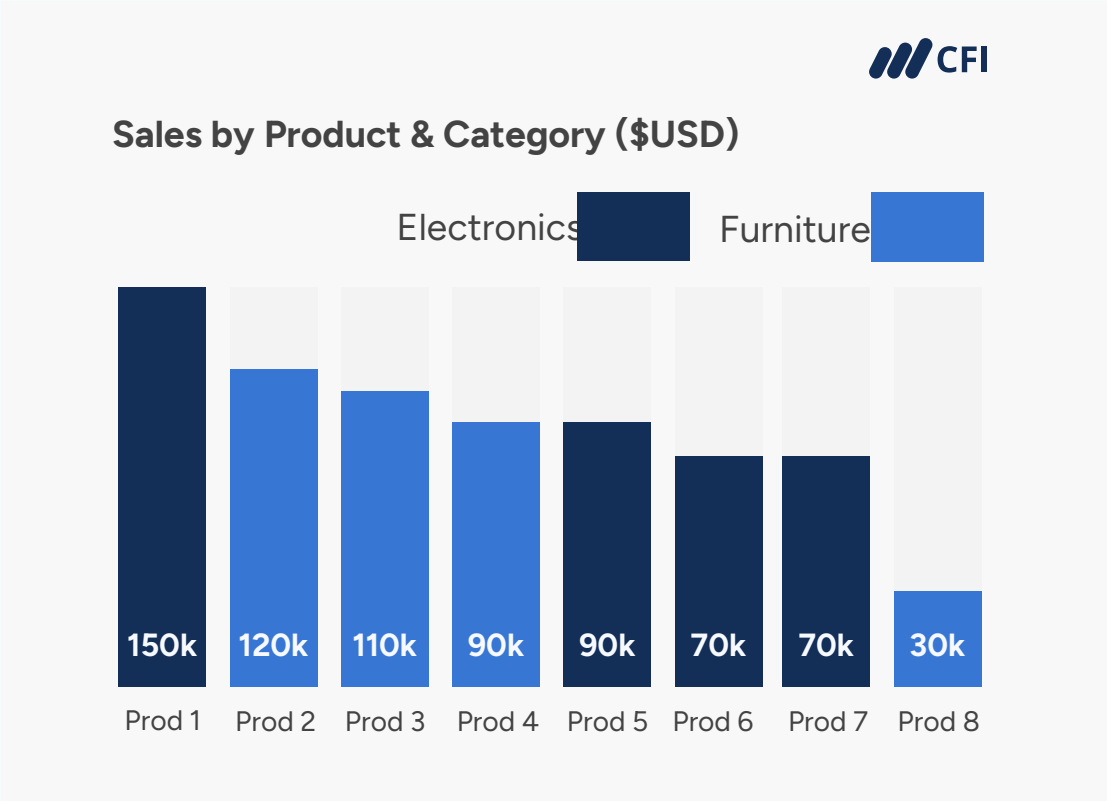


Good use of color to indicate association.

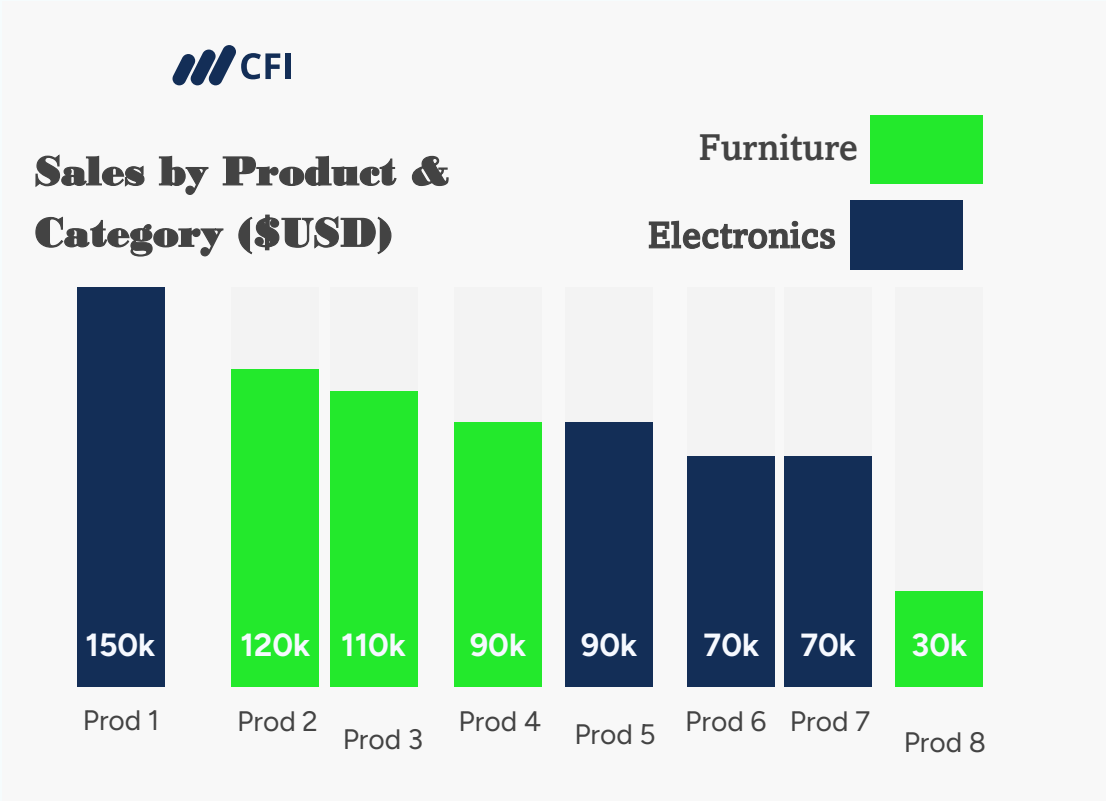


Color used with no intent.

Design & Aesthetics



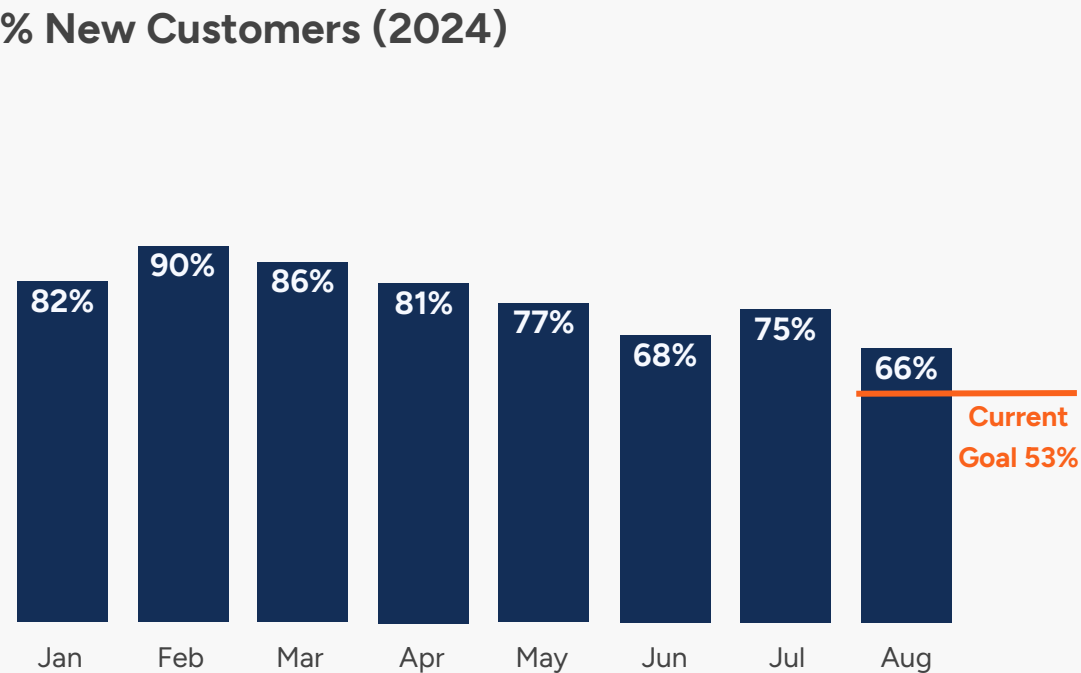
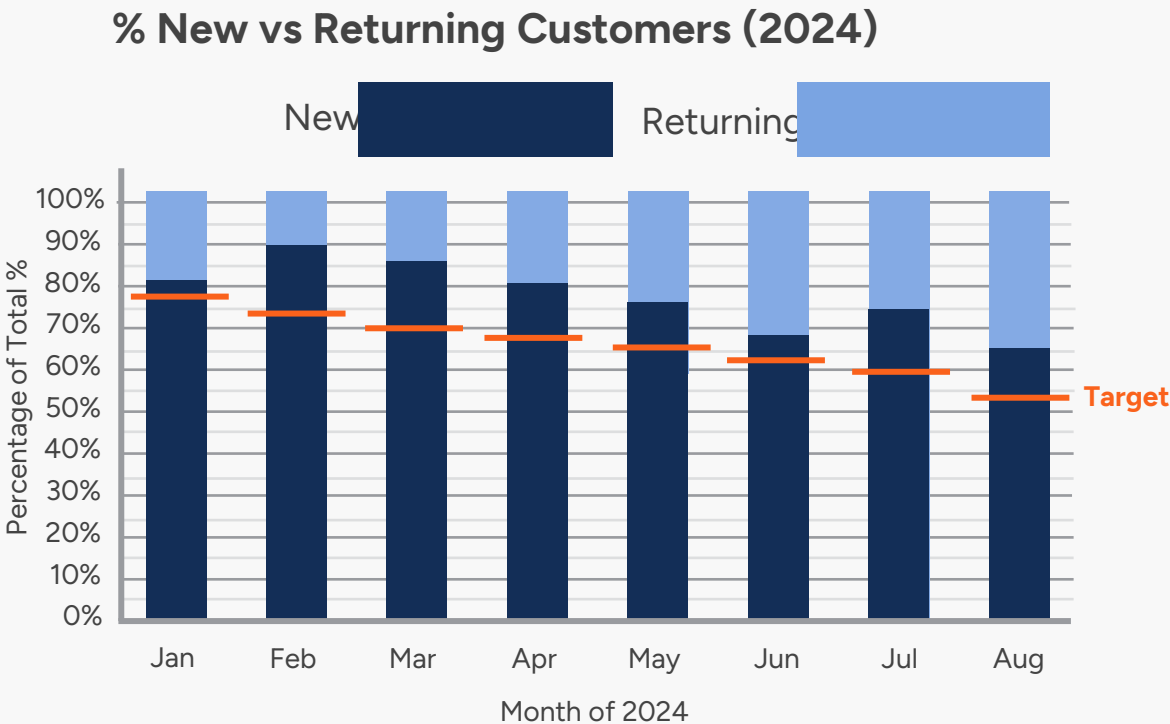
- ✓ Alignment is good.
- ✓ Fonts are consistent.
- ✓ Brand colors match the CFI logo.



- ✗ Bad alignment is distracting..
- ✗ Fonts are not consistent, which looks unprofessional.
- ✗ Green color does not match the brand.

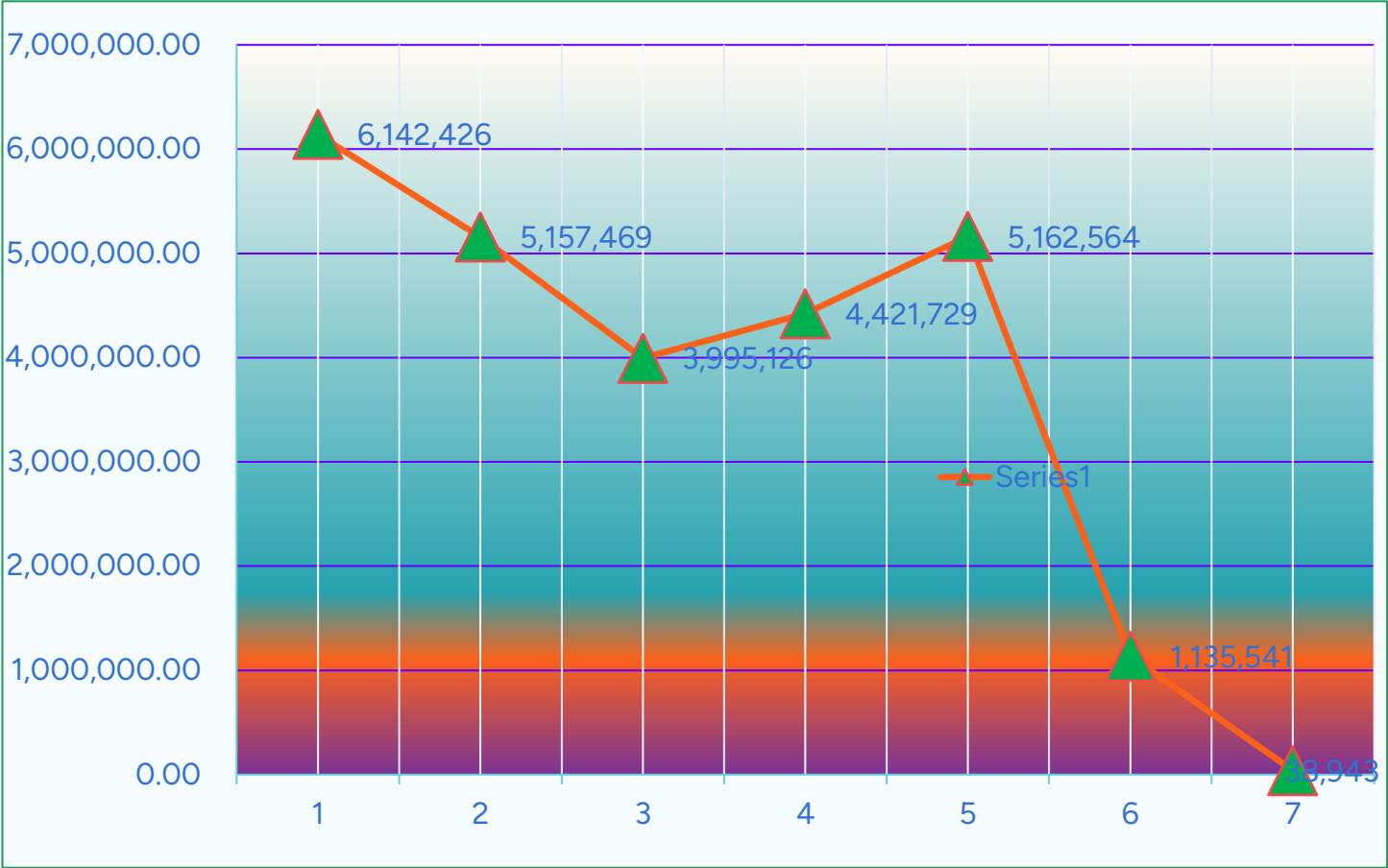
Design & Aesthetics

More pixels and more features on our charts can make them **overwhelming**.



Removing **gridlines & features**, and **increasing white space** is one of many easy ways to make **simpler, more impactful charts**.

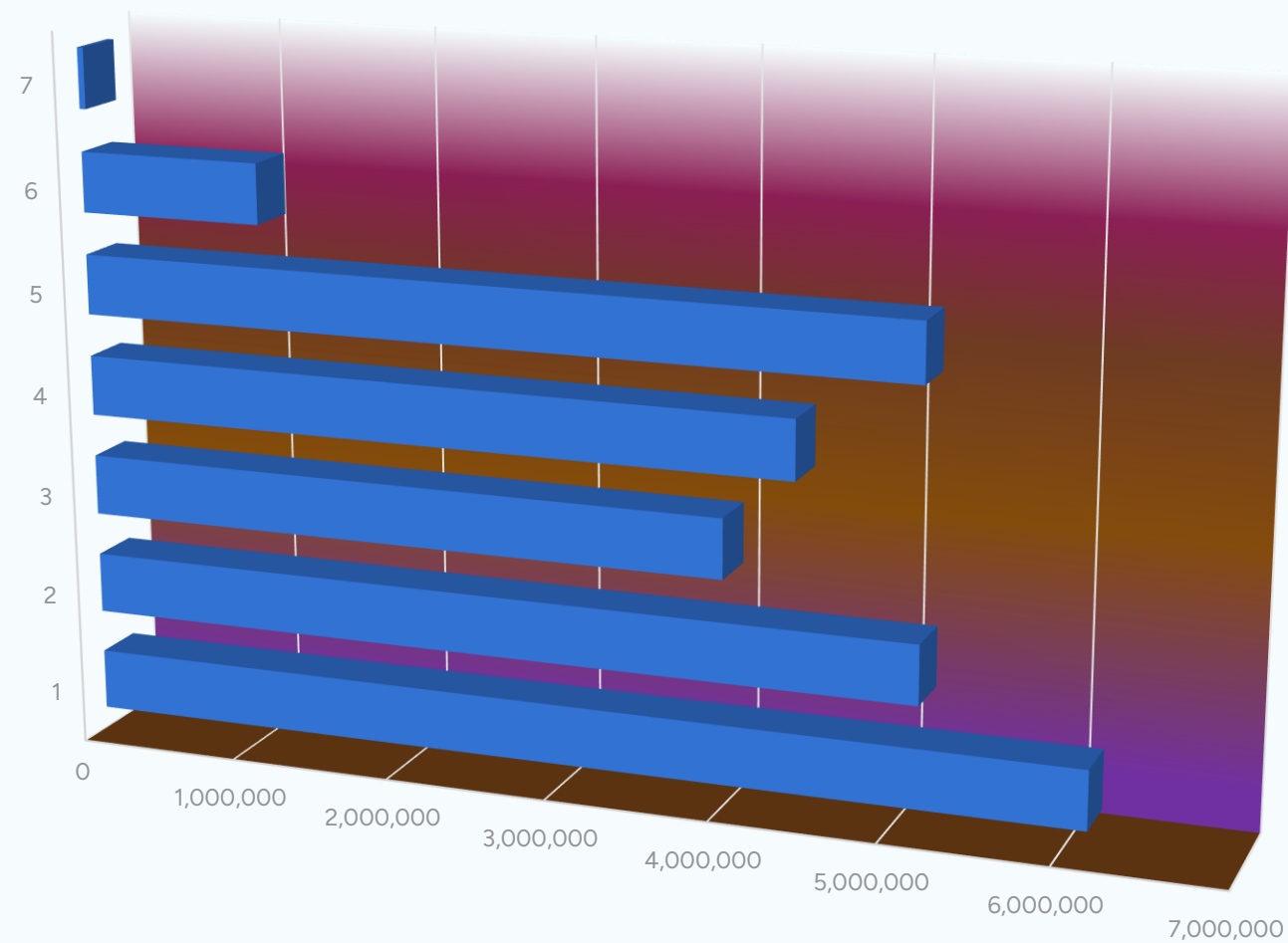
What Not to Do



What Not to Do



What Not to Do



Dashboarding

Importance of Decision-Making

Learn to think like a business leader and anticipate their next question.

Help your leaders understand what decisions to make and how to improve performance.

- So what? How does this help me take action?
- What are the key takeaways or most important numbers?
- What should I focus my attention on?
- Why? What can we do about it?
- Where can I get that information?
- What are the root causes?
- What can we do to fix the issue?

Planning & Process

A few steps to increase your chances of success...

Is it necessary at all?

- Avoid dashboard overload.
- Is a dashboard the most effective solution?

Define Your Audience

- Who wants the dashboard and why?
- Have a clear stakeholder for each dashboard.
- Be accountable for the time invested in developing it.

Define Your Focus Question

- What are you trying to answer?
- Use this to stay on track and avoid scope creep.

The Bigger Picture

- Why is this question and dashboard important?
- What are the wider business implications?
- What are the likely next questions?

Planning & Process

A few steps to increase your chances of success...

Exploration

- You may need to do some exploratory analysis to understand the data.

Narrative & Visuals

- Plan the narrative of your dashboard.
- What's the main message or metrics?
- Map out the sequence of visuals and charts.

Context

- Are you communicating the relevant supporting information?
- What assumptions have been made?
- What is the relevant time period?
- Does this represent the whole business or just part of it?
- When was the data updated?
- Does the data fall within expected values? If not, why not?

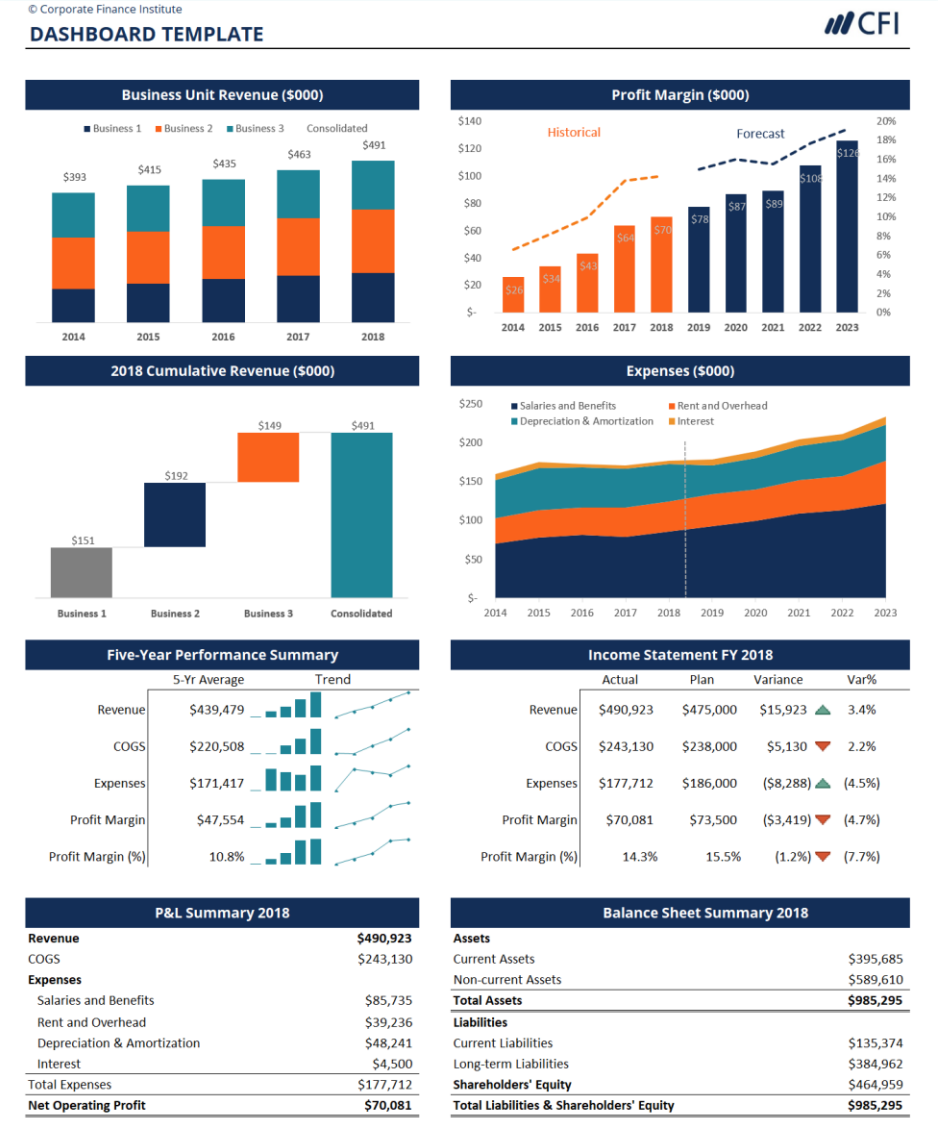
Planning & Process

A few steps to increase your chances of success...

Testing

- Organize opportunities to put your dashboard in front of its audience.
- Get feedback on what works and what doesn't.

Dashboarding Tools



Adoption

A few thoughts to help you gain adoption with your dashboards.

Key Points

- Establish a senior leadership champion who will actively use and show it in meetings.
- Keep stakeholders and champions up to date and ensure that the dashboard works for them.
- Does your dashboard drive actionable insights to important questions?